

Week 02. Distribution, Variance and its applications

Part 2: Statistics and Econometrics. MK68-2023/24

Eduard Bukin

eduard.bukin@agrar.uni-giessen.de

Justus Liebig University Giessen

Institute of Agricultural Policy and Market Research

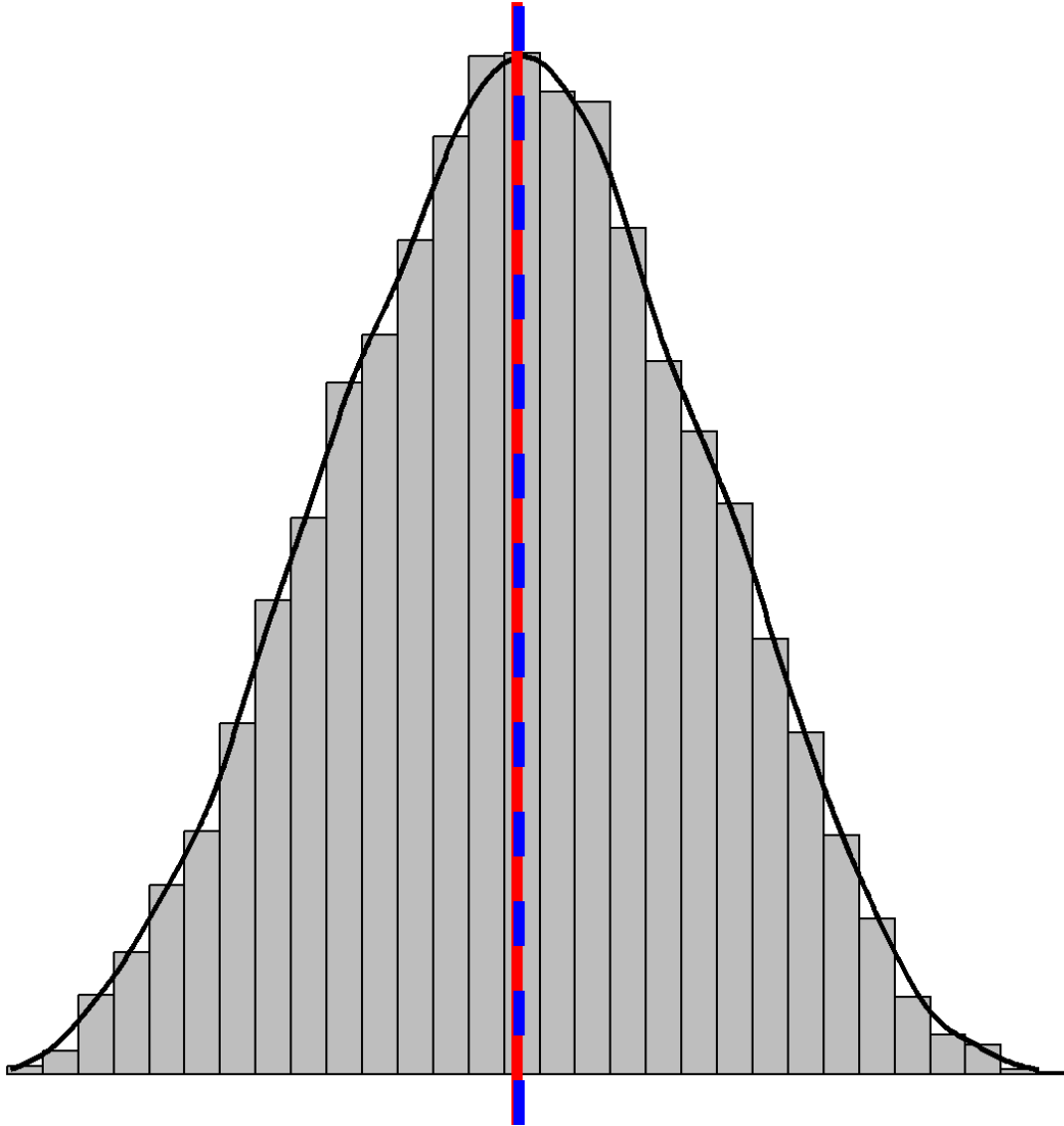
Content

- Distribution
- Cumulative Distribution
- Cumulative Distribution: applications
- Variance
- Quantiles
- Application of variance in the descriptive analysis
- The (Standard) Normal Distribution
- When univariate data analysis is not enough?
- Takeaways checklist
- References

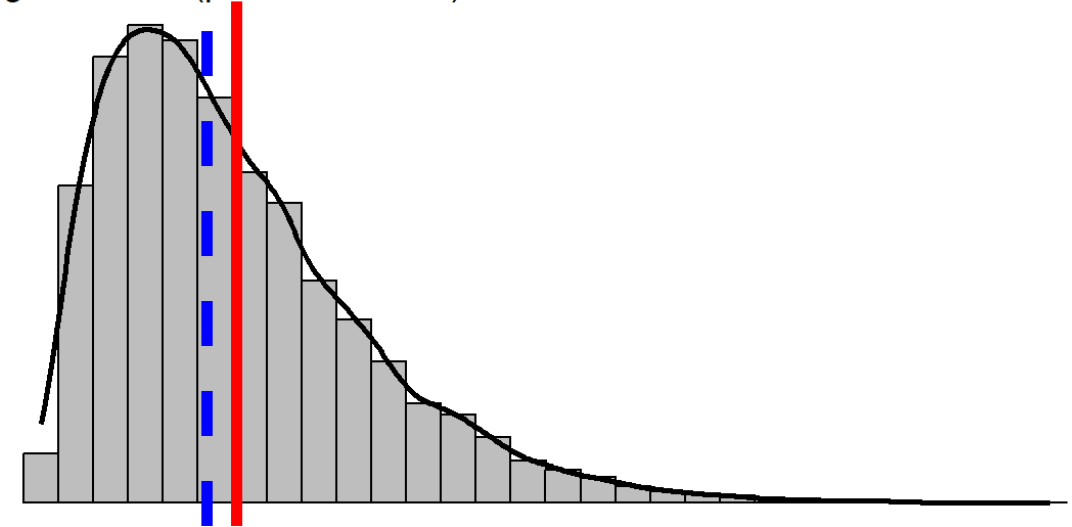
Distribution

Symmetric vs Non symmetric

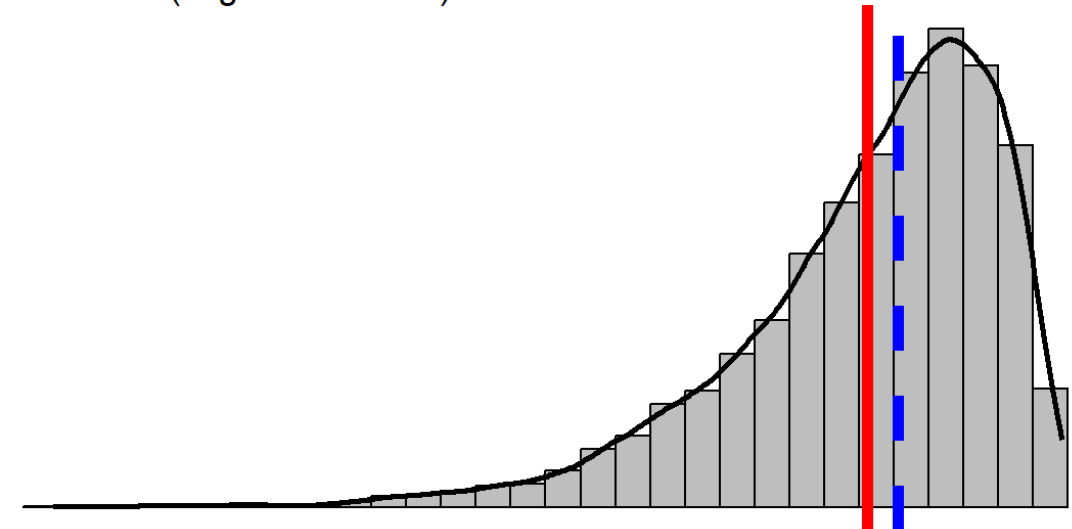
Symmetric distribution



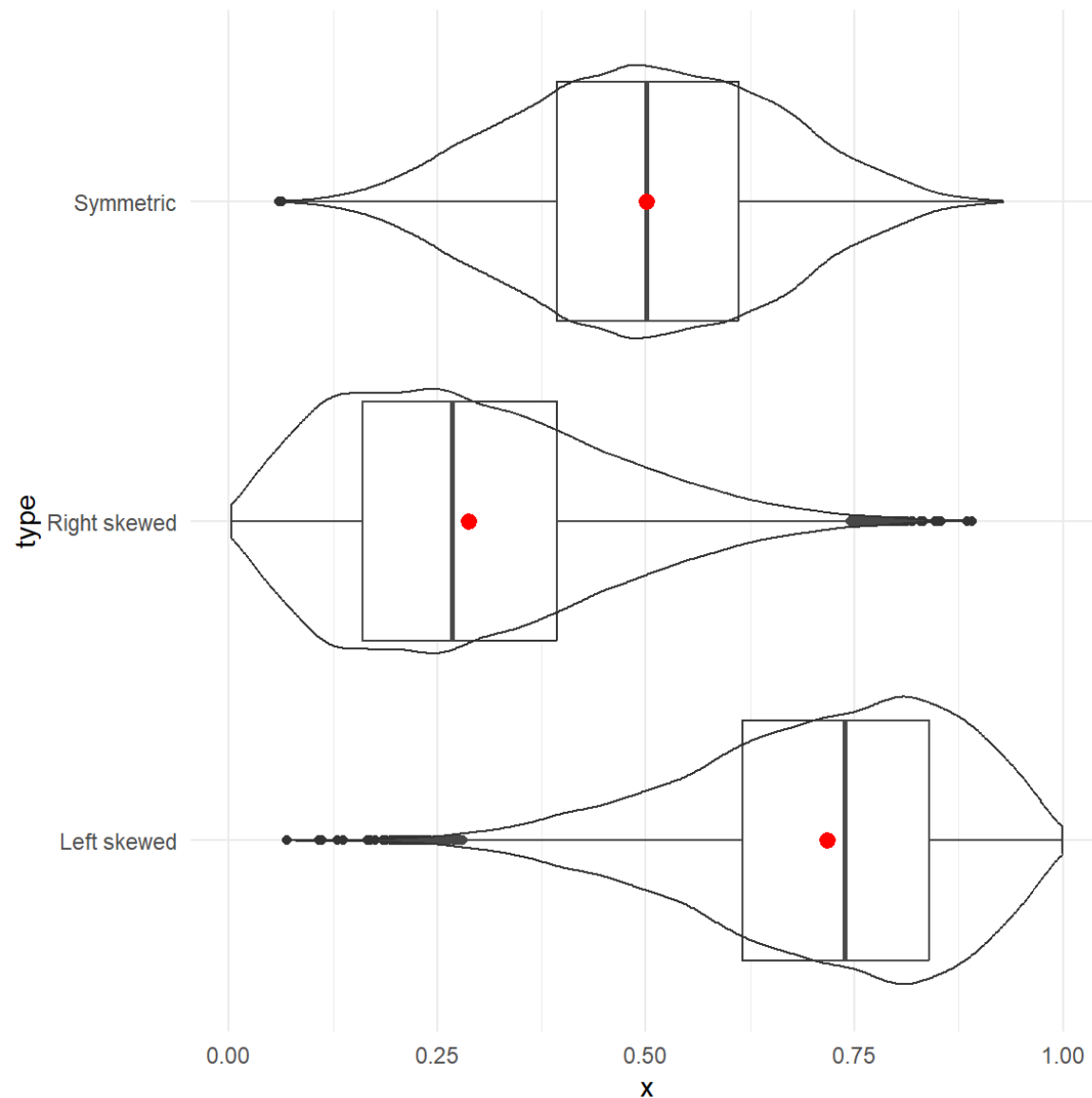
Right skewed (positive skewed)



Left skewed (negative skewed)



Extremem and influential observations



- Mean is heavily affected by influential observations.
- Median is more **robust**.

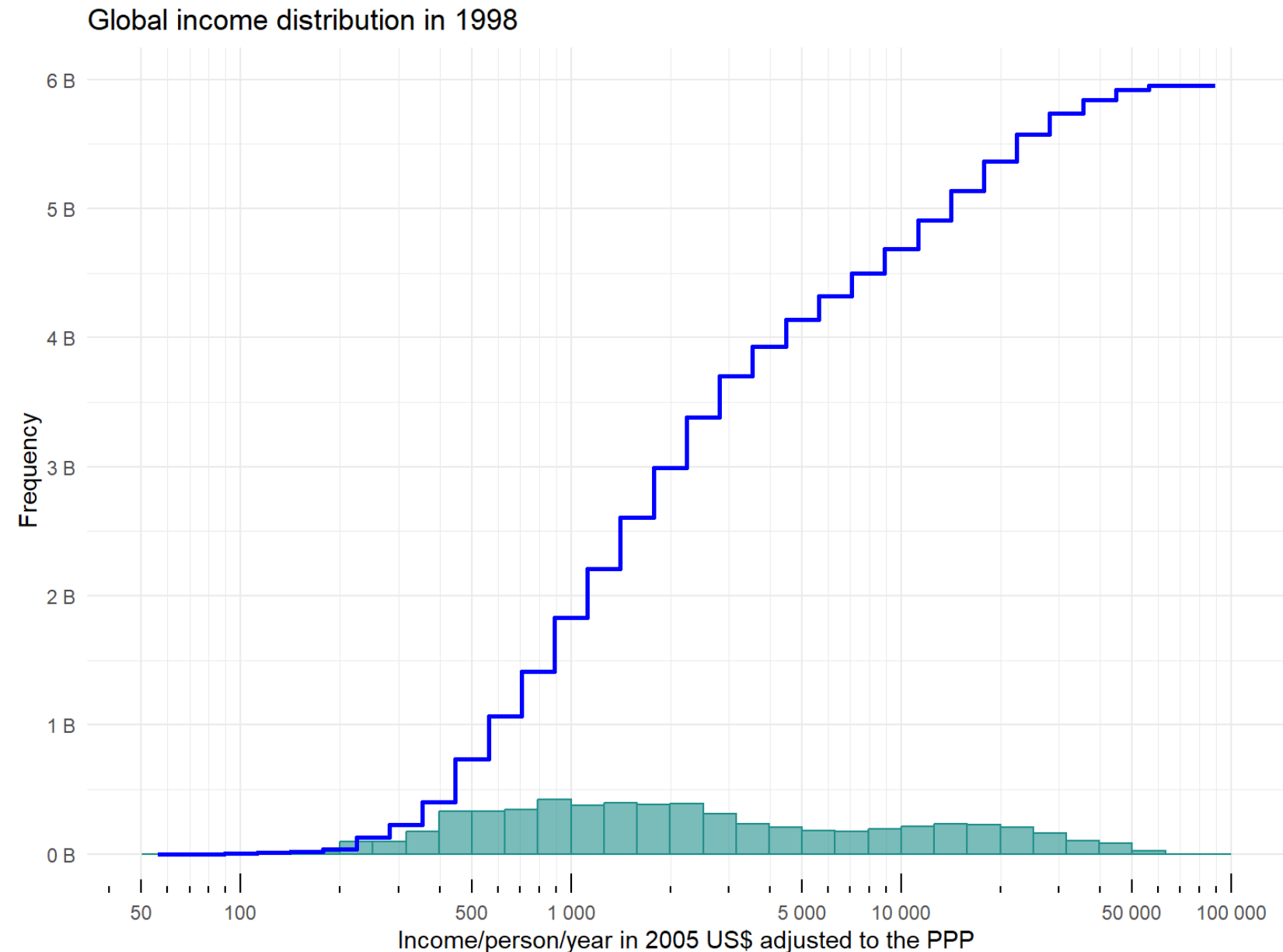
Cumulative Distribution

- Show all the data points at once with their contribution to the distribution of the value x
- See [Ch. 8 Visualizing distributions: Empirical cumulative distribution functions and q-q plots](#) in Fundamentals of Data Visualization ([Wilke 2019](#))

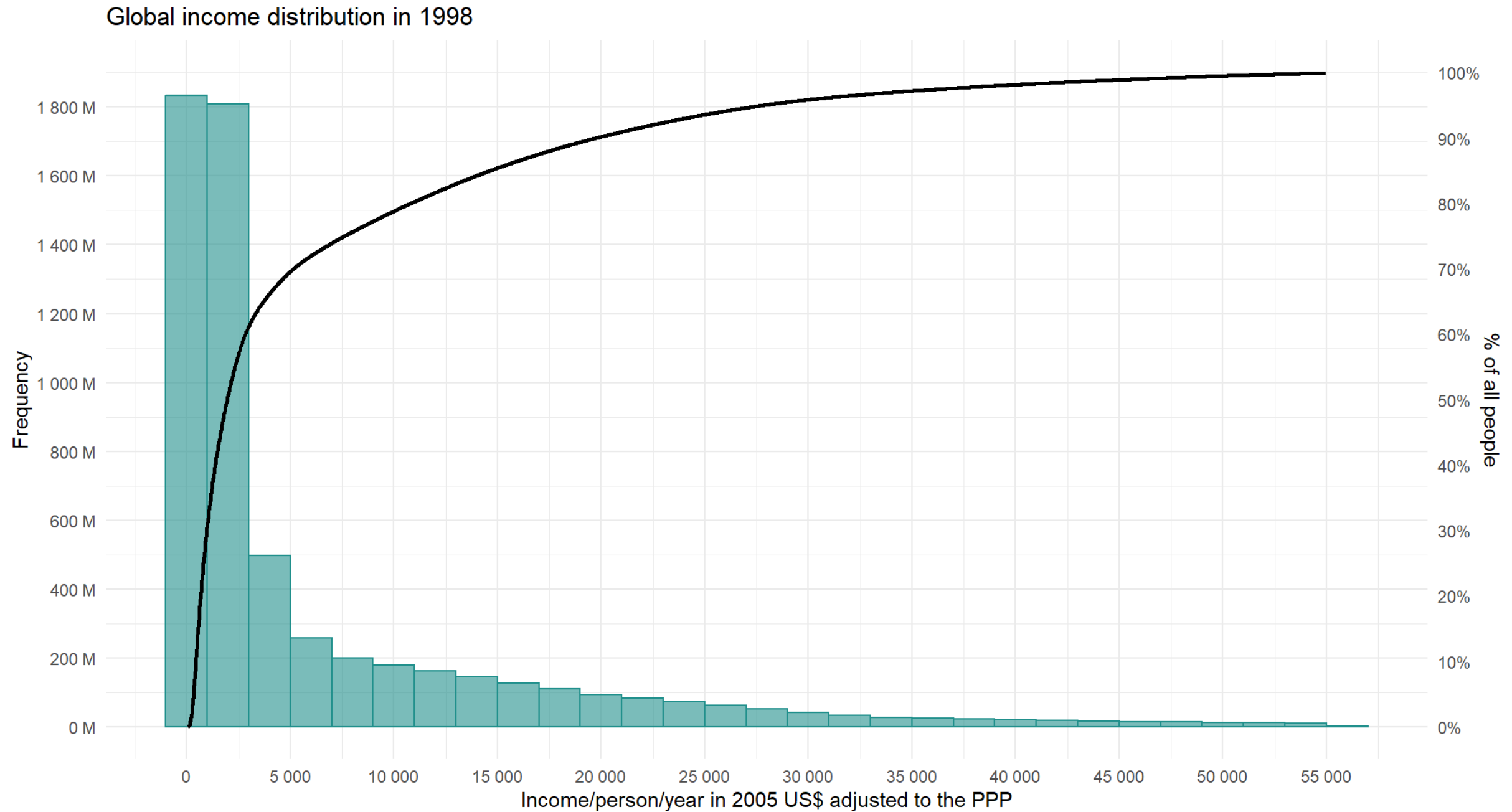
From a density to a cumulative distribution

How to:

- Order all observation ascendingly.
- Plot cumulative frequency for every existing observation on Y axis.

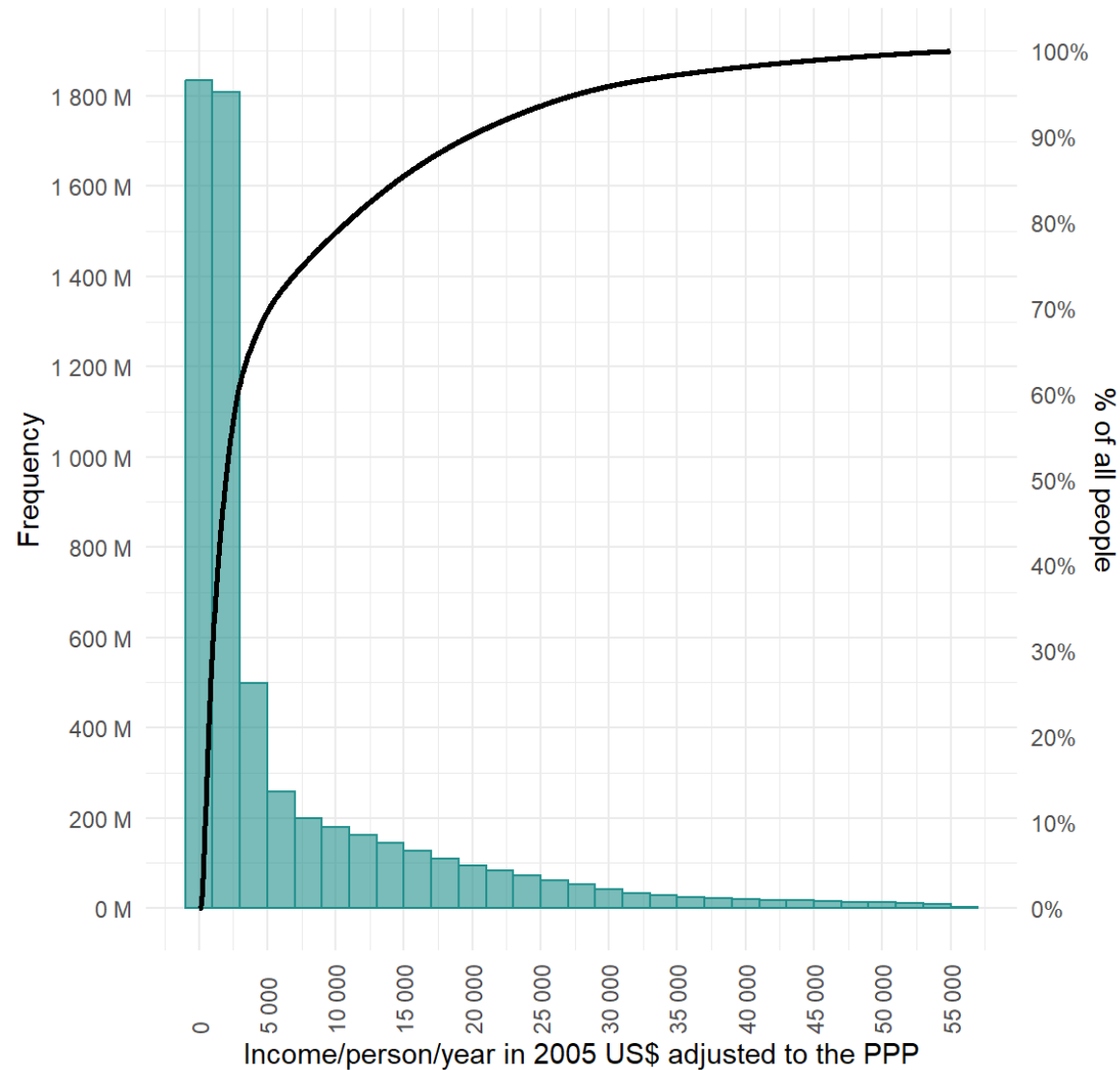


Cumulative distribution of income (1)

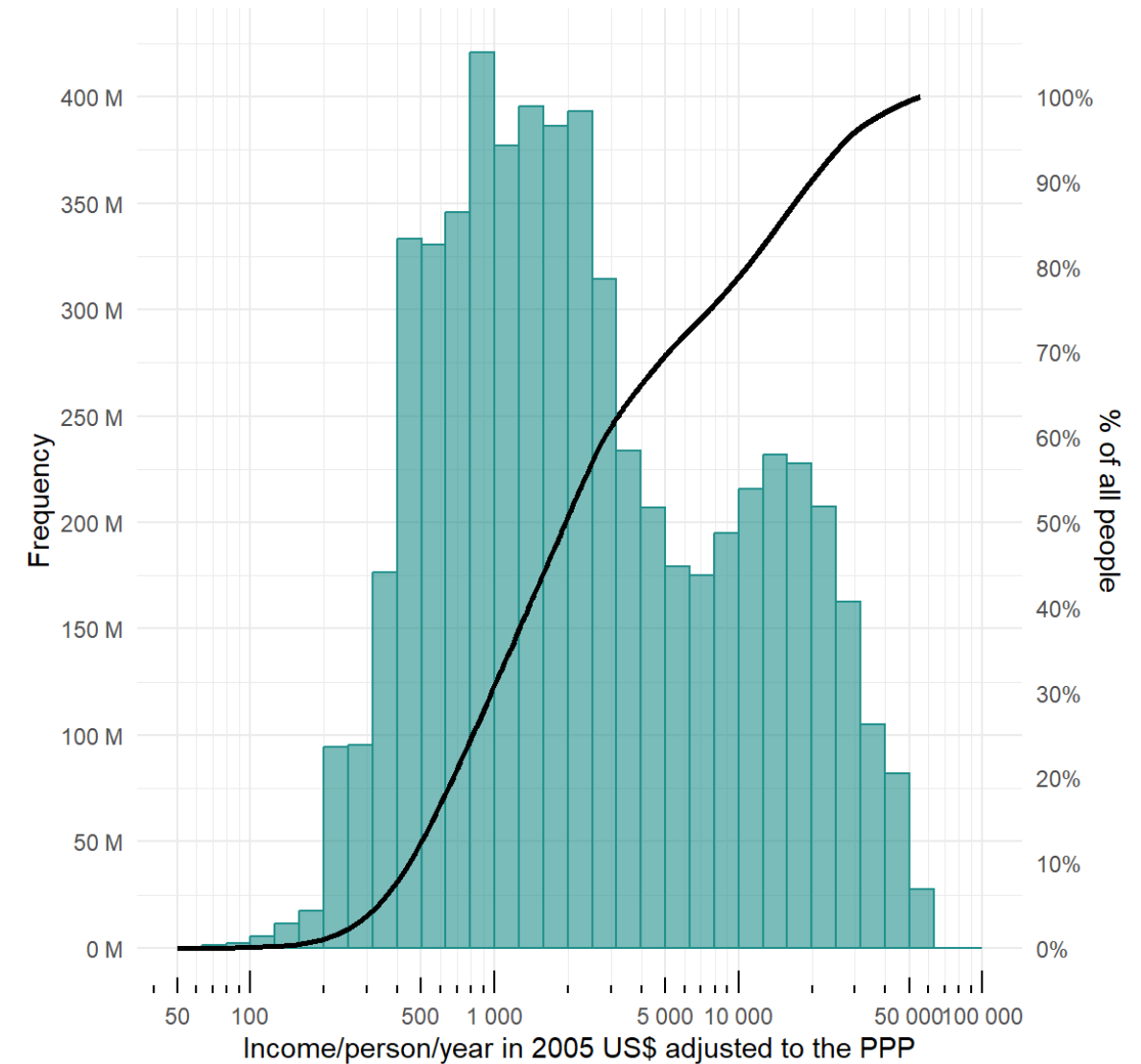


Income distribution

Global income distribution in 1998



Global income distribution in 1998

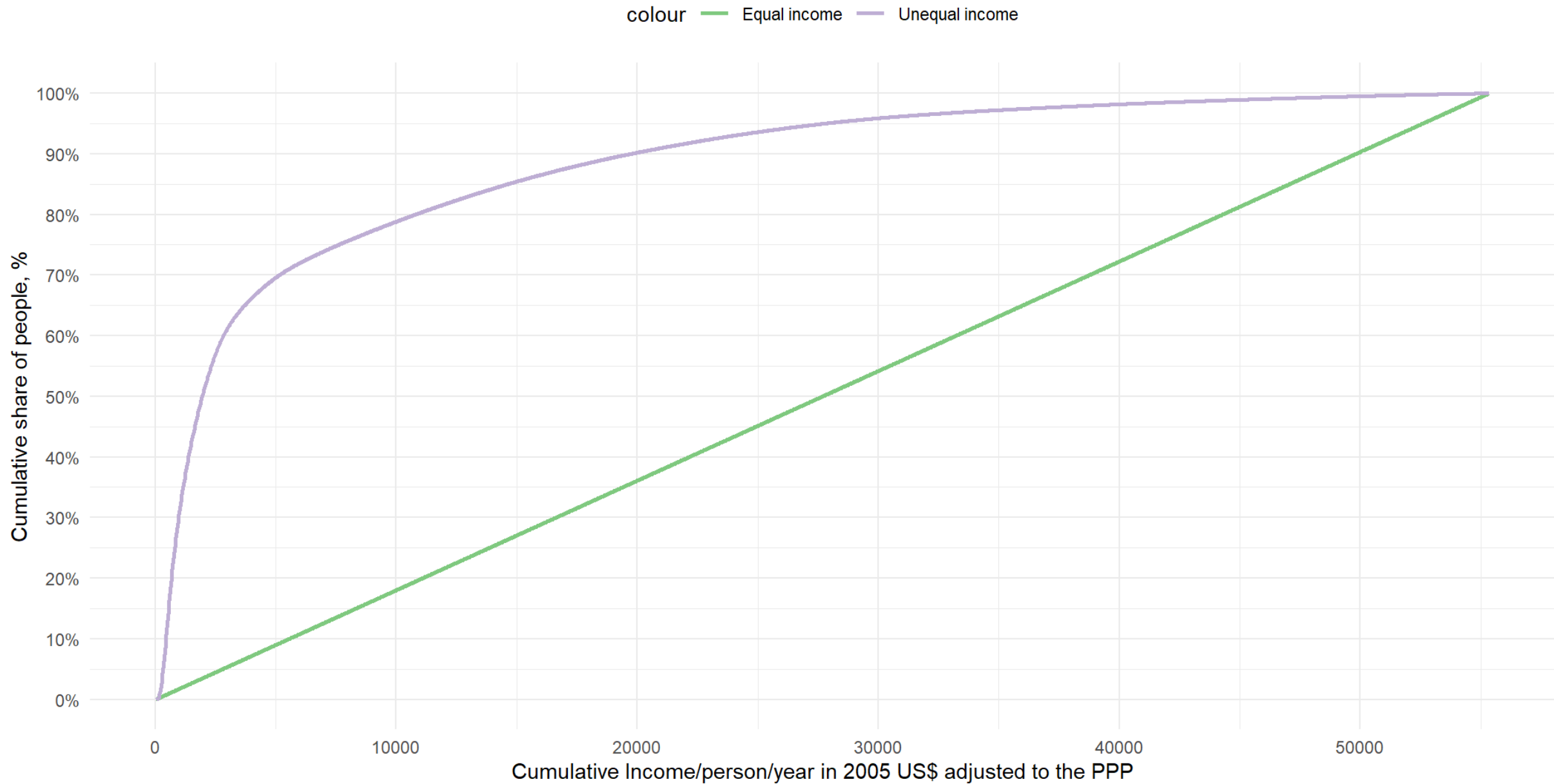


Cumulative Distribution: applications

- Measuring income inequality

Income inequality

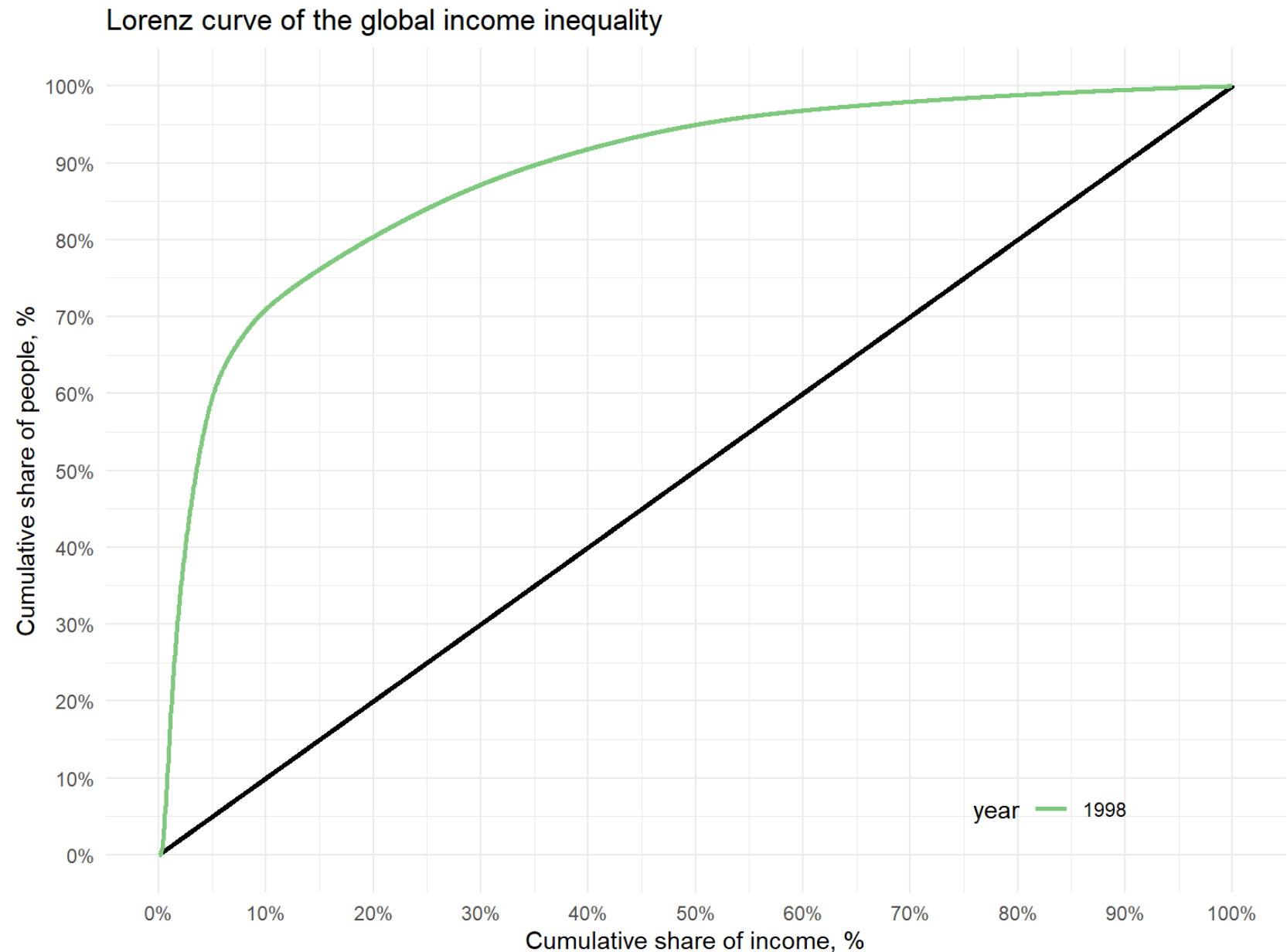
Equally vs unequally distributed income



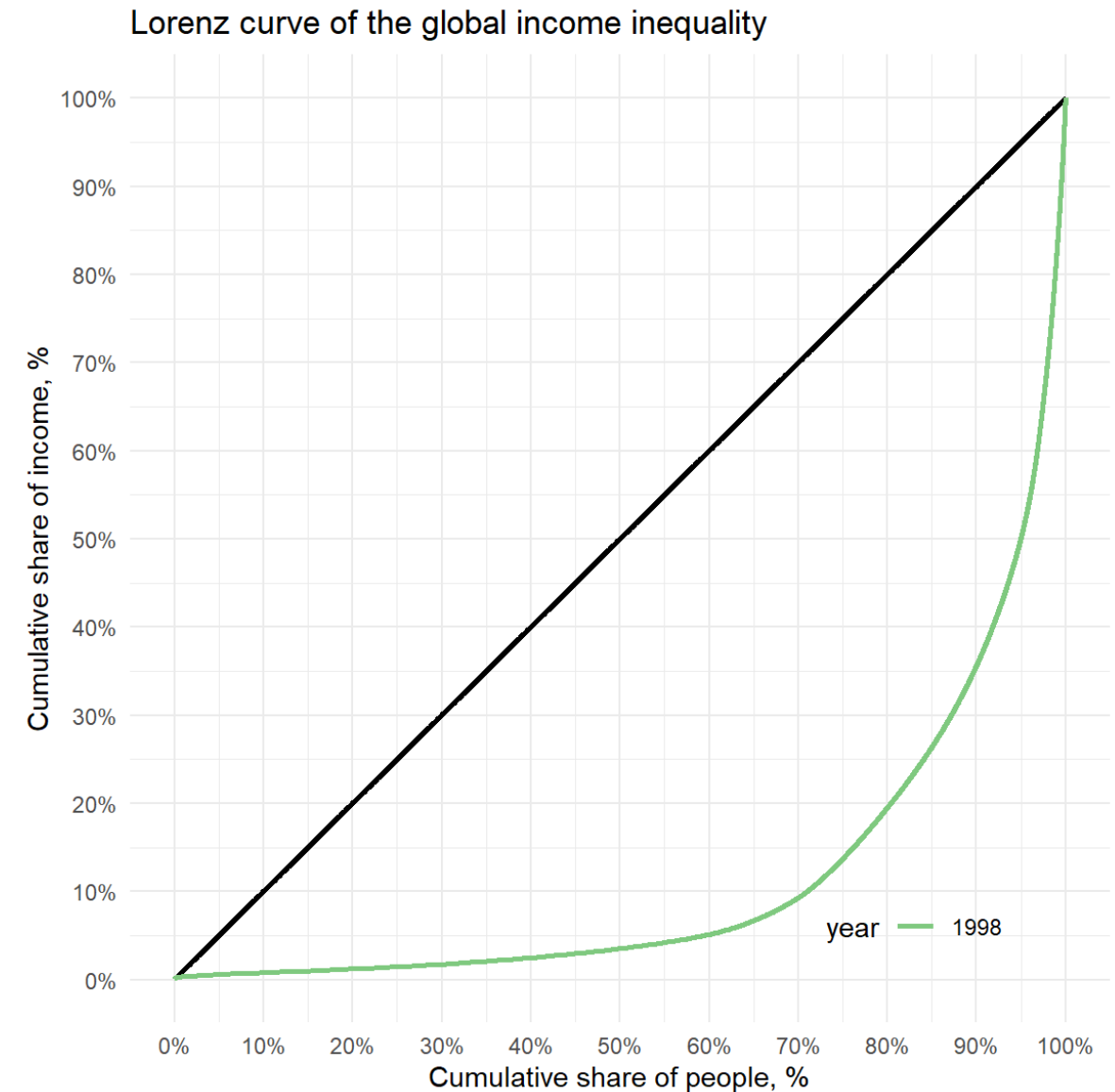
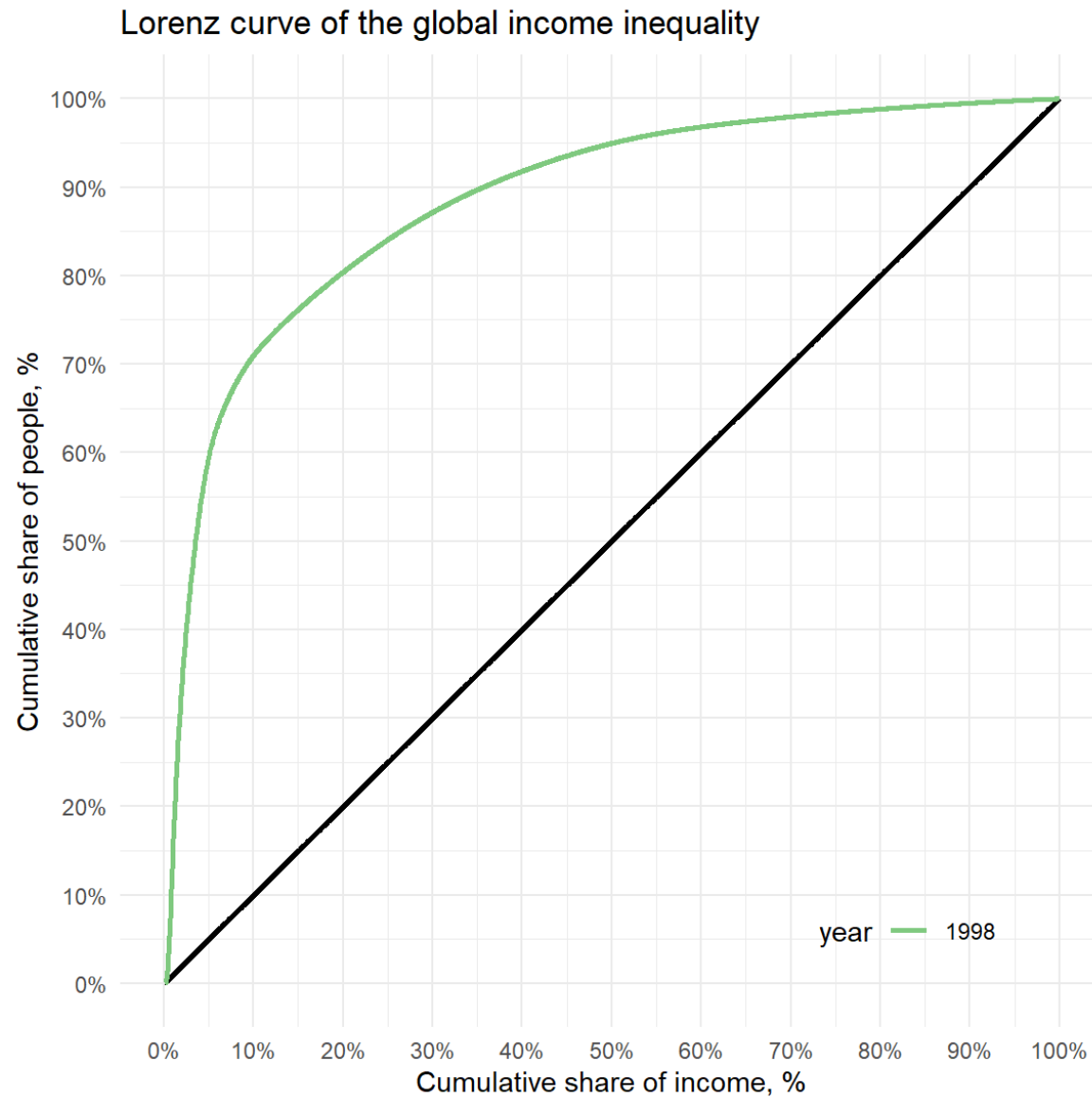
Lorenz curve

The [Lorenz curve](#):

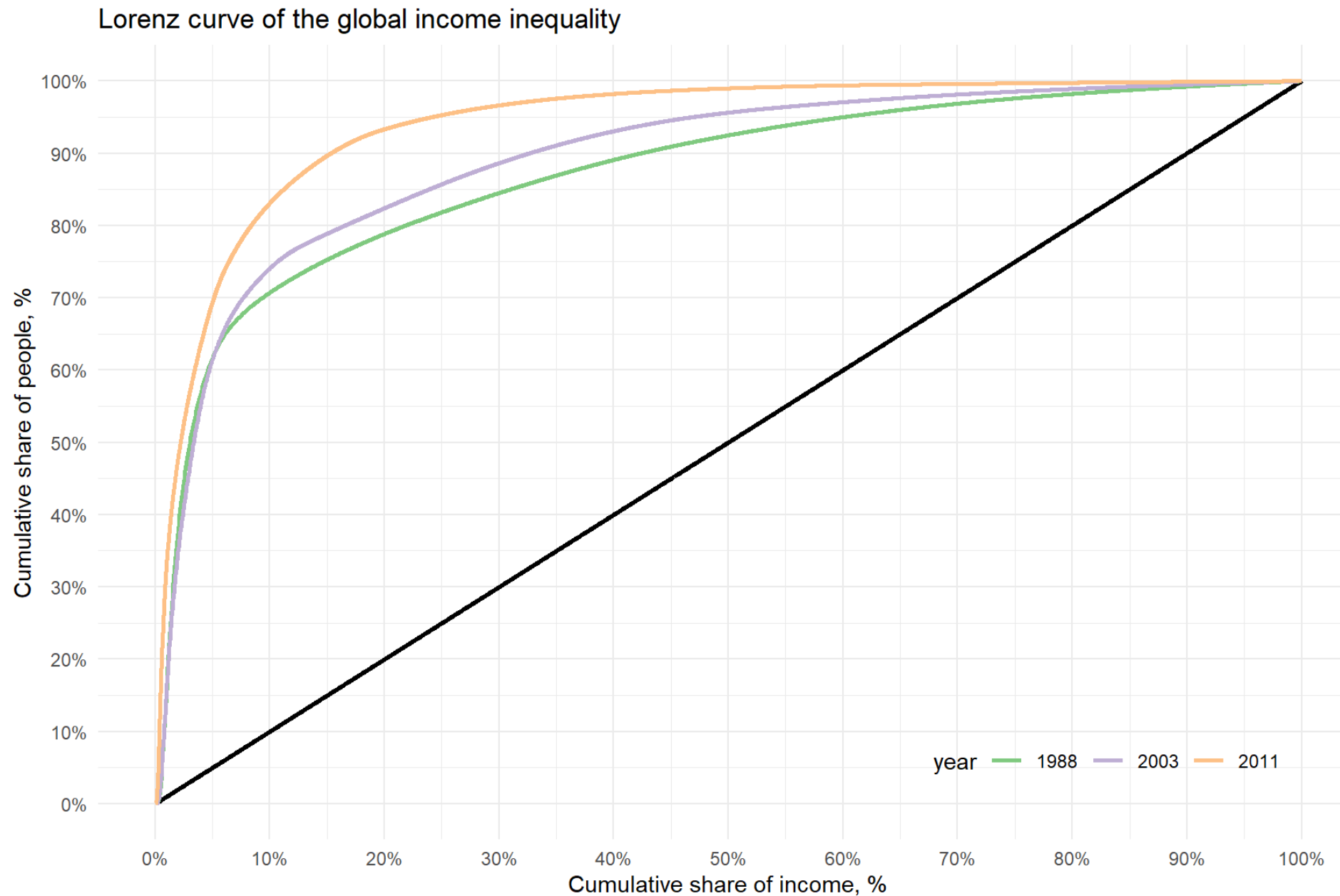
- It is a cumulative distribution function.
- Arrange all individuals by increasing income
 - from 0% to 100% of maximum income.
- Compute cumulative % of people by income level (0 to 100% of all people).
- see [Graphical representation of the Lorenz curve](#).



Lorenz curve: spot any differences

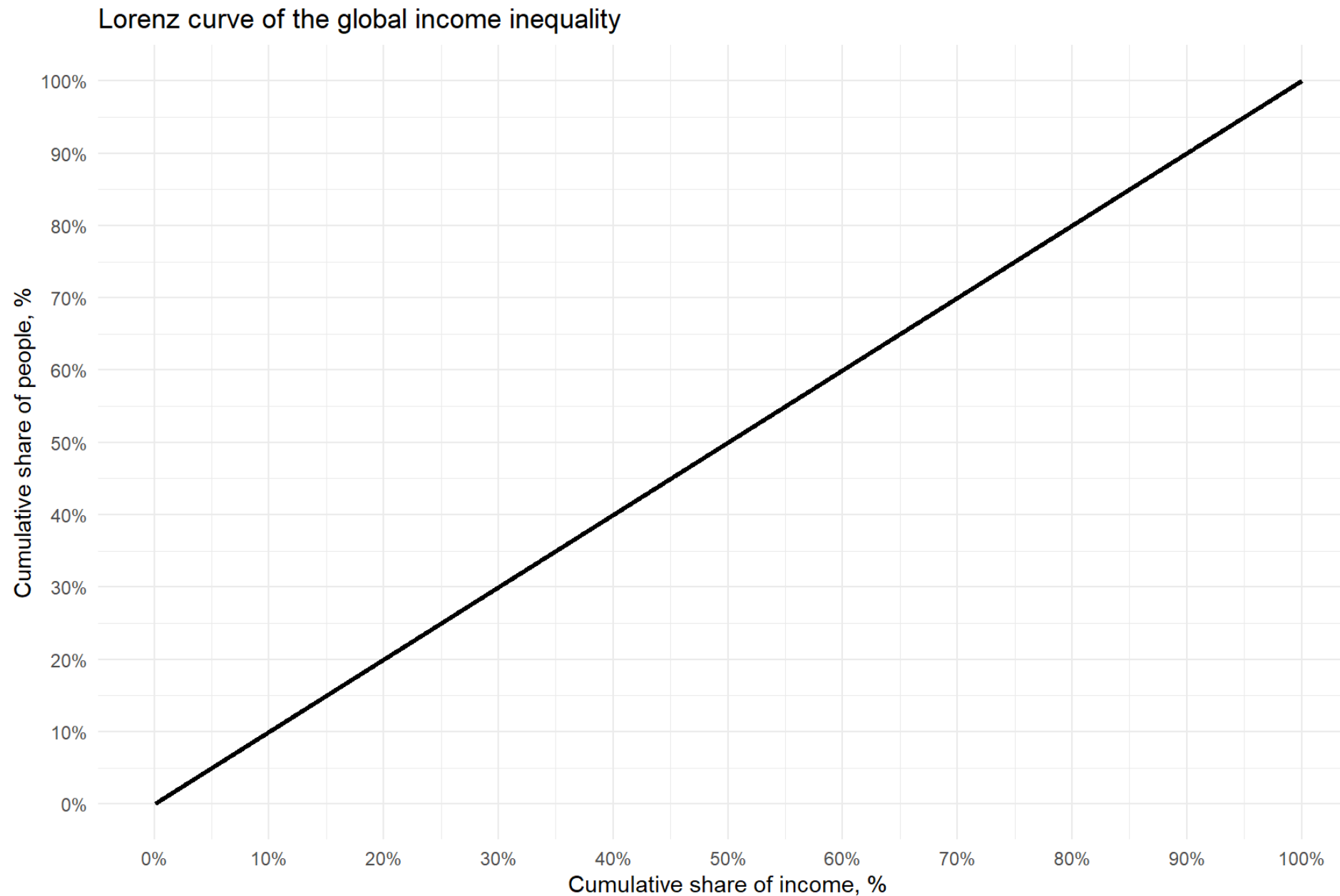


Exam problems example: Lorenz curve



- How has income distribution changed between 1988 and 2011?
- What statistical concept is used in the backbone of the Lorenz curve computation?

Lorenz curve: comparing distributions drawing



- Draw two Lorenz curves both with unequal income distribution.
- Explain differences between two curves?

At home: Read what is the Gini Coefficeint

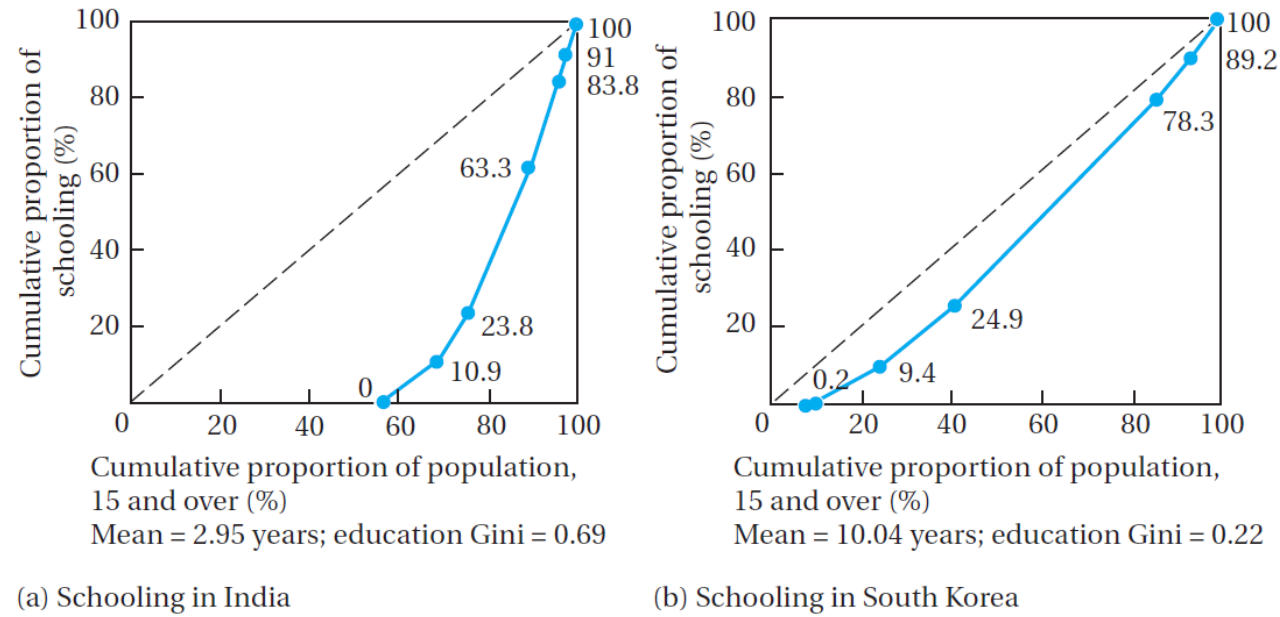
Joe Hasell (2023) - “Measuring inequality: What is the Gini coefficient?”

Published online at OurWorldInData.org. Retrieved from:

<https://ourworldindata.org/what-is-the-gini-coefficient> [Online Resource]

At home: Lorenz curve: Inequality

FIGURE 8.7 Lorenz Curves for Education in India and South Korea



Source: From *The Quality of Growth*. Copyright © 2000 by World Bank. Reprinted with permission.

- What kind of plot is this?
- What does it show?
- Which of these two countries demonstrate more equal distribution?
- Why do you conclude this?

Variance

- **Variance** ($Var[X]$) is a measure of the spread of a distribution relative to its expected value (Dowdy, Weardon, and Chilko 2004)
 - is the measure of variability in a single variable;
 - is the average squared distance from the mean;
 - is in squared units (**has no intuitive meaning**);
- **Standard Deviation (SD)** is the squared root of the variance:
 - $SD = \sqrt{\text{Variance}}$
 - The standard deviation is
 - **in the same units of measurement as the measured variable**

Variance

Population variance, σ^2

$$\sigma^2 = \frac{\sum_{i=1}^N [x_i - \mu]^2}{N},$$

where μ is the population mean:

$$\mu = \sum_{i=1}^N x_i / N$$

Only used with the population

Standard Deviation:

$$\sigma = \sqrt{\sigma^2}$$

Sample variance, s^2

$$s^2 = \frac{\sum_{i=1}^N [x_i - \bar{x}]^2}{N - 1},$$

where $\bar{x}$ is the sample mean:

$$\bar{x} = \sum_{i=1}^N x_i / N$$

 Most commonly used in statistics

Sample Standard Deviation:

$$s = \sqrt{s^2}$$

Example: the variance estimation

id	x	$x - \bar{x}$	$abs(x - \bar{x})$	$(x - \bar{x})^2$
1	10.9	-0.6672	0.6672	0.4452
2	9.92	-1.6472	1.6472	2.7133
3	26.3	14.7328	14.7328	217.0554
4	9.92	-1.6472	1.6472	2.7133
5	9.43	-2.1372	2.1372	4.5676
...	...	...	...	...
7	17.09	...	...	30.5013
		5.5228	5.5228	

Sum: $\sum x = 578.36$

N. obs: $N = 50$

Sample mean:

$$\bar{x} = \sum x / N$$

$$\bar{x} = 11.57$$

Mean absolute deviation:

$$MAD = \sum |x - \bar{x}| / N$$

$$MAD = 3.93$$

$$\text{Sum: } \sum (x - \bar{x})^2 = 1250.67$$

Sample variance:

$$s^2 = \sum (x - \bar{x})^2 / (N - 1)$$

$$s^2 = 25.52$$

Standard Deviation:

$$s = \sqrt{25.52} = 5.05$$

Standard Deviation

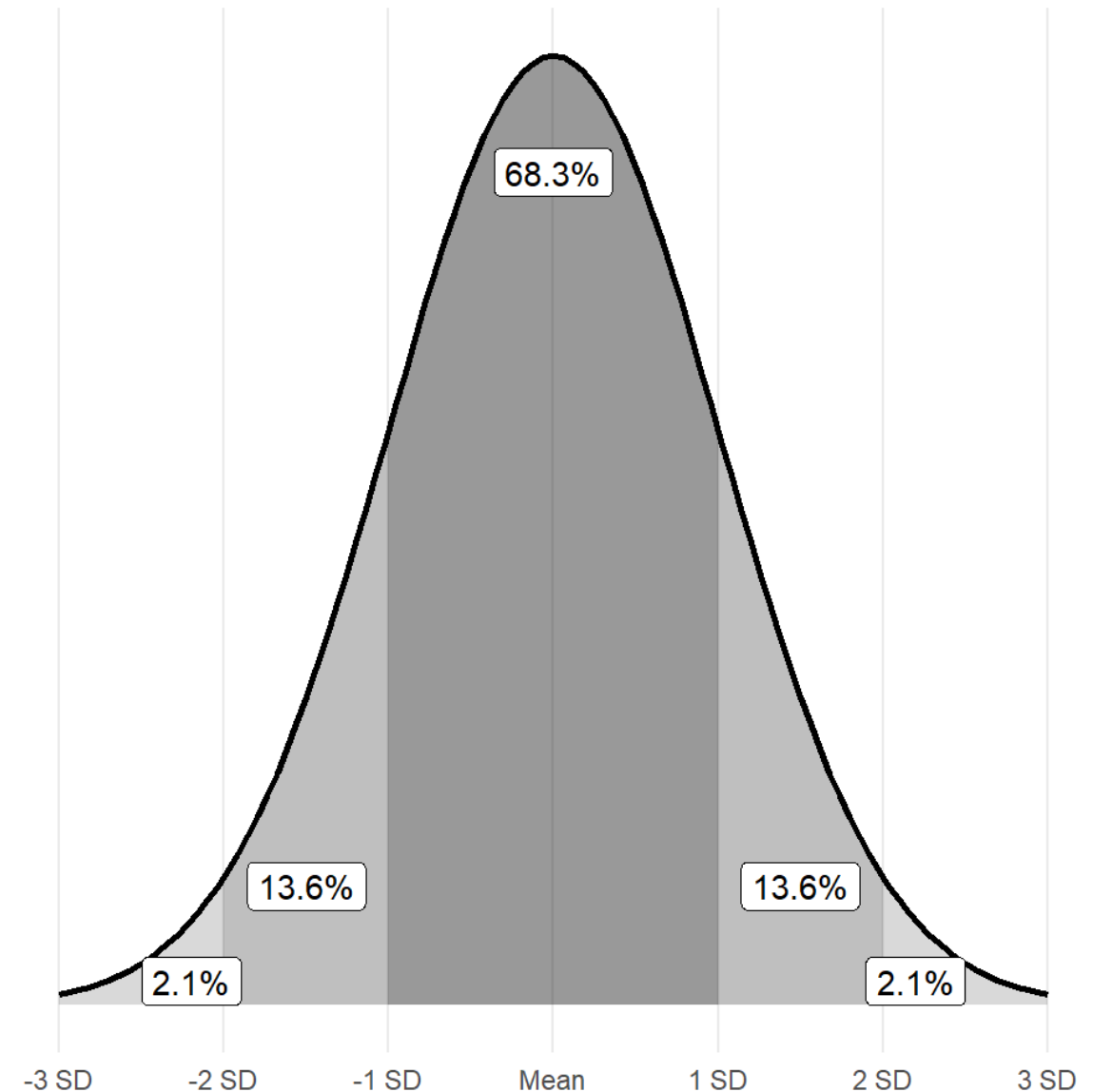
represents the typical deviation of observations from the mean.

If the distribution is normal:

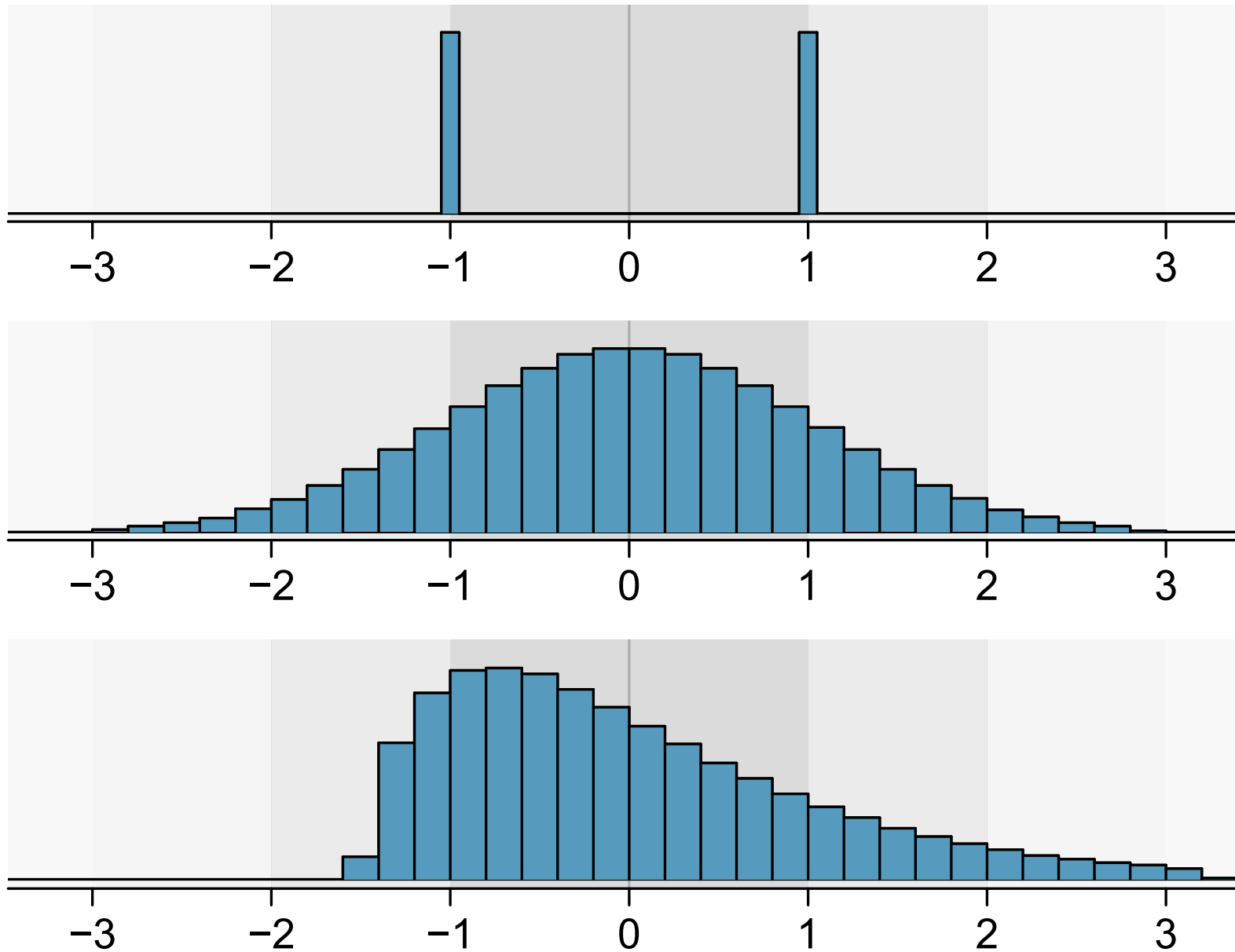
- 68.3% obs. are in ± 1 SD;
- 95.4% obs. are in ± 2 SD;
- 99.7% obs. are in ± 3 SD;



The standard deviation is in the same units of measurement as the measured variable.



Not every distribution is normal



i Tip!

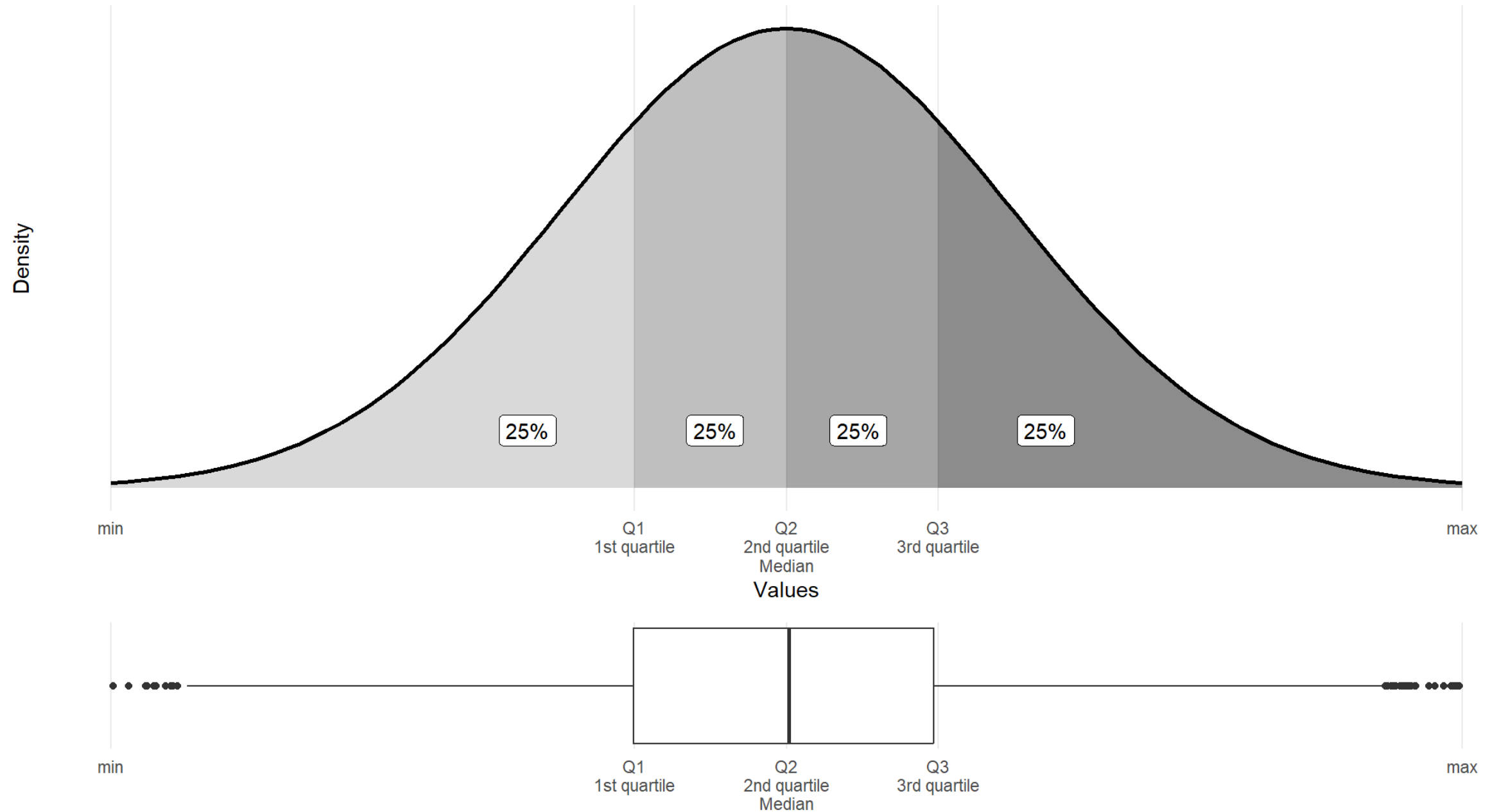
A good description of distribution should include

- Central tendency;
- SD (Variance);
- Description of the shape;
 - modes
 - symmetry/skew

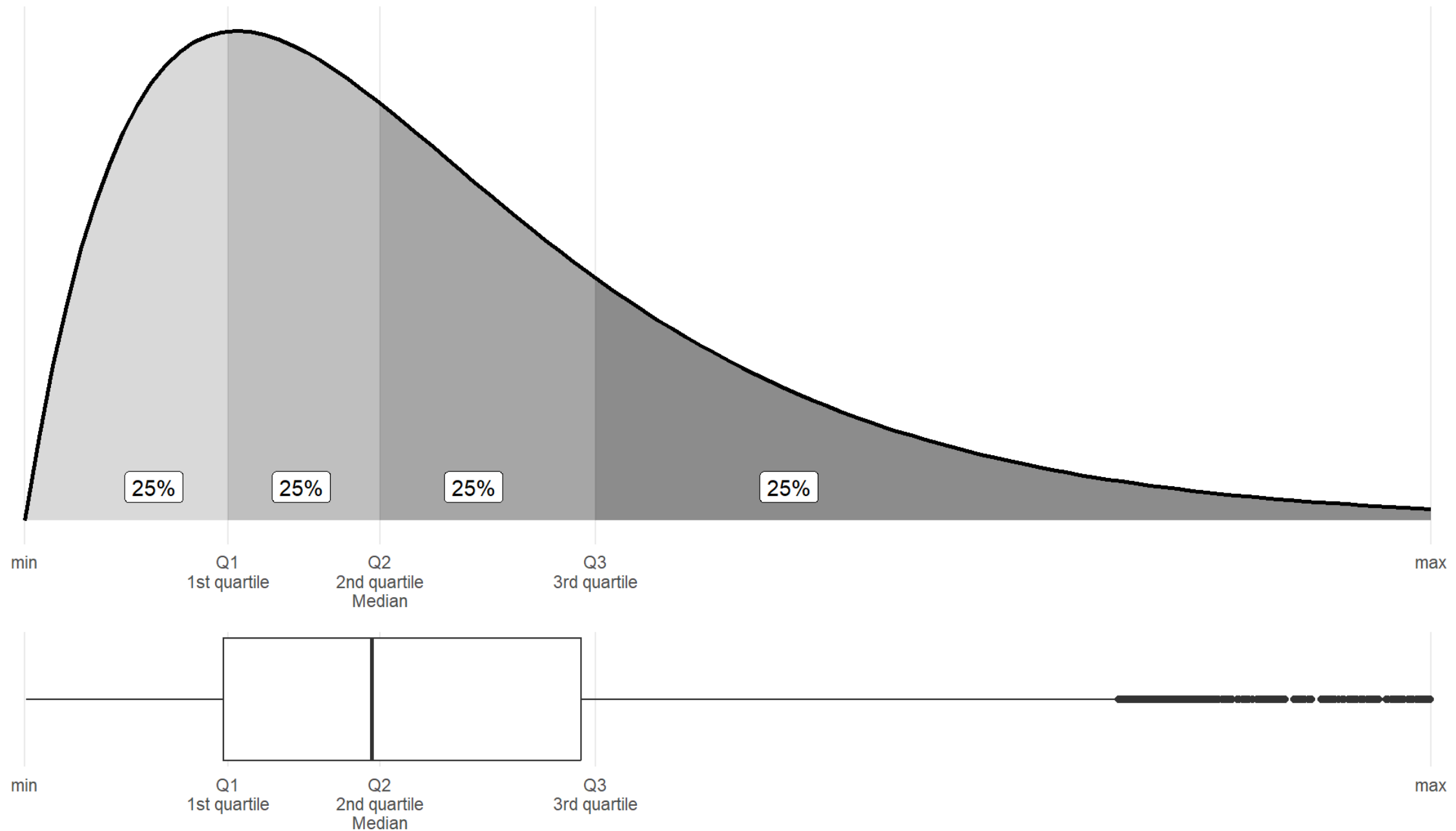
Quantiles

- cut points that divide our data into equal chunks with equal probability of being selected (equal number of observations in each chunk).
- The Robust measures of the variance that depends on the quantities of the ordered observations rather than their values:
- Quantiles: Quartiles, Deciles, Percentiles...
 - see [Quantiles and Percentiles, Clearly Explained](#)
- Range
- IQR - Interquartile range

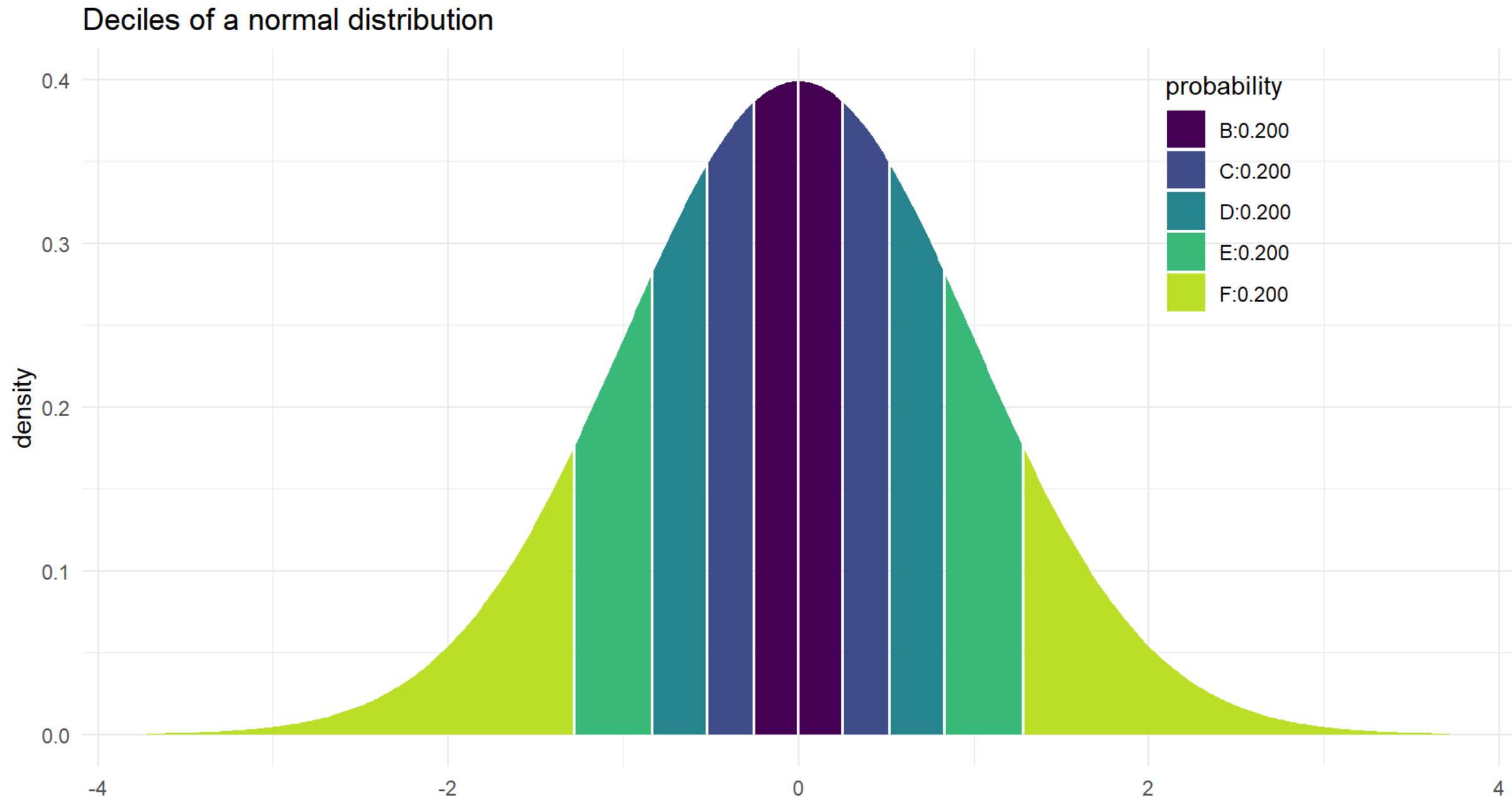
Quartiles: every 25% of observations



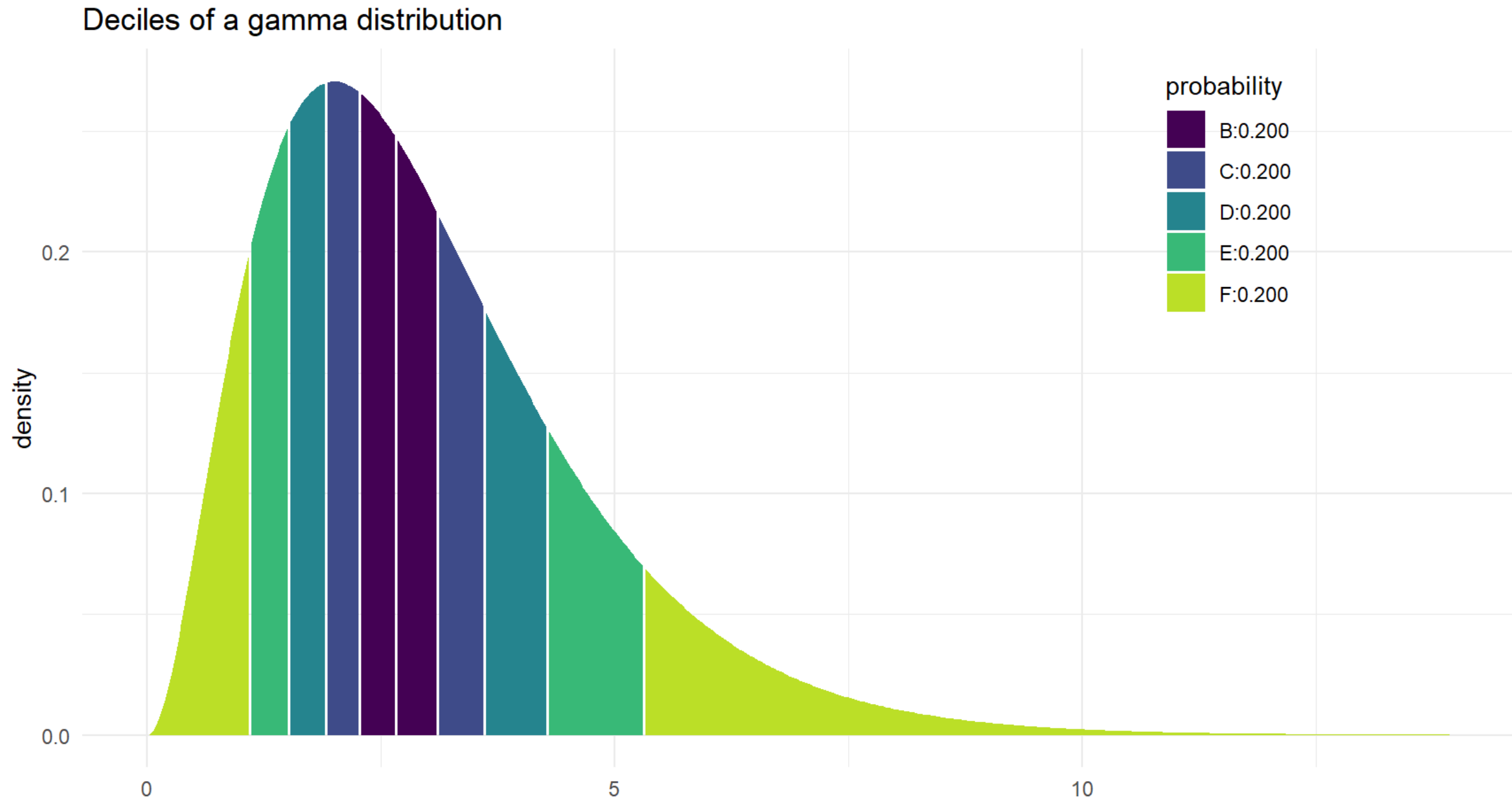
Quartiles: non-normal distribution



Deciles: every 10% of observations (Normal)

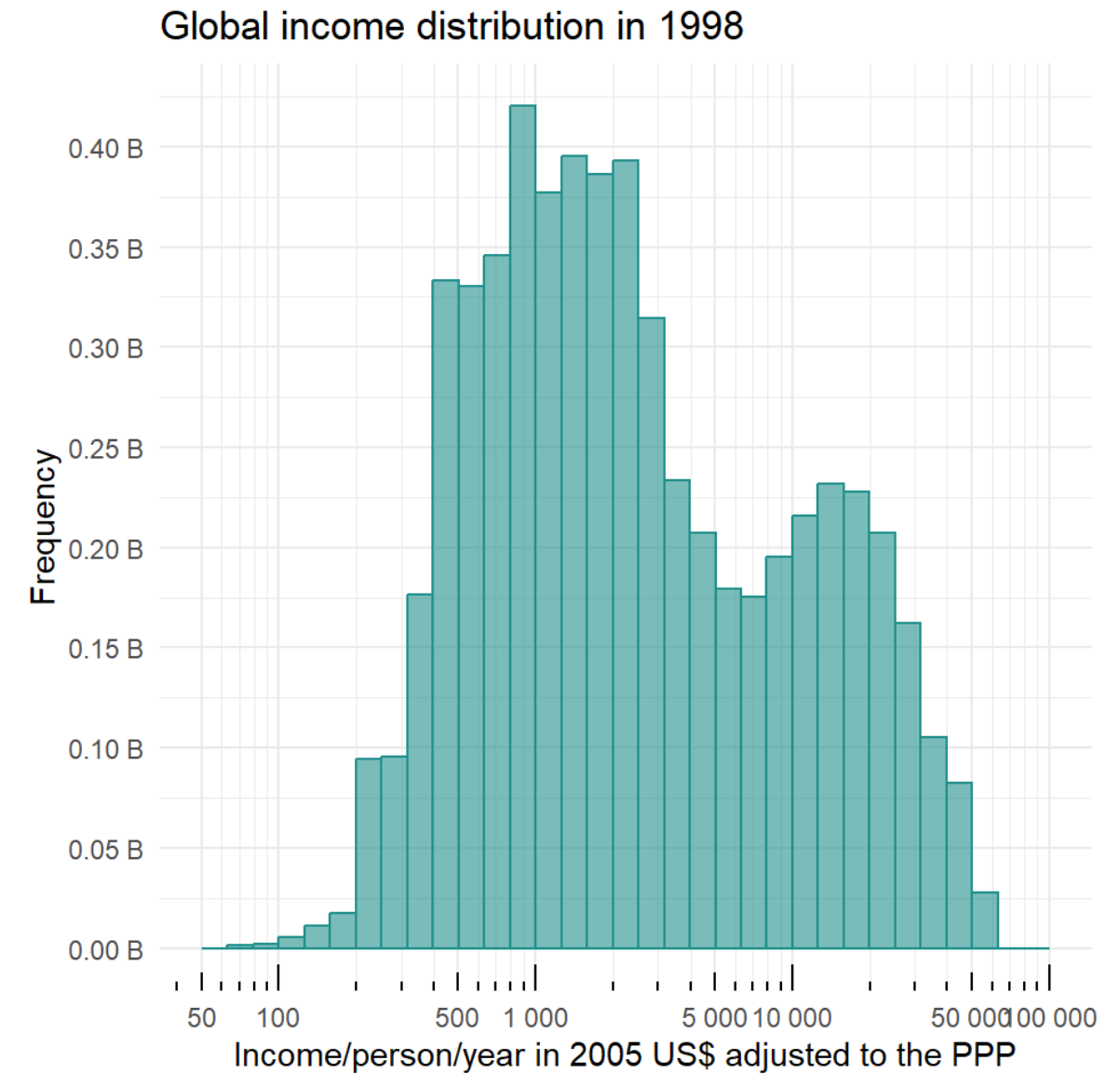
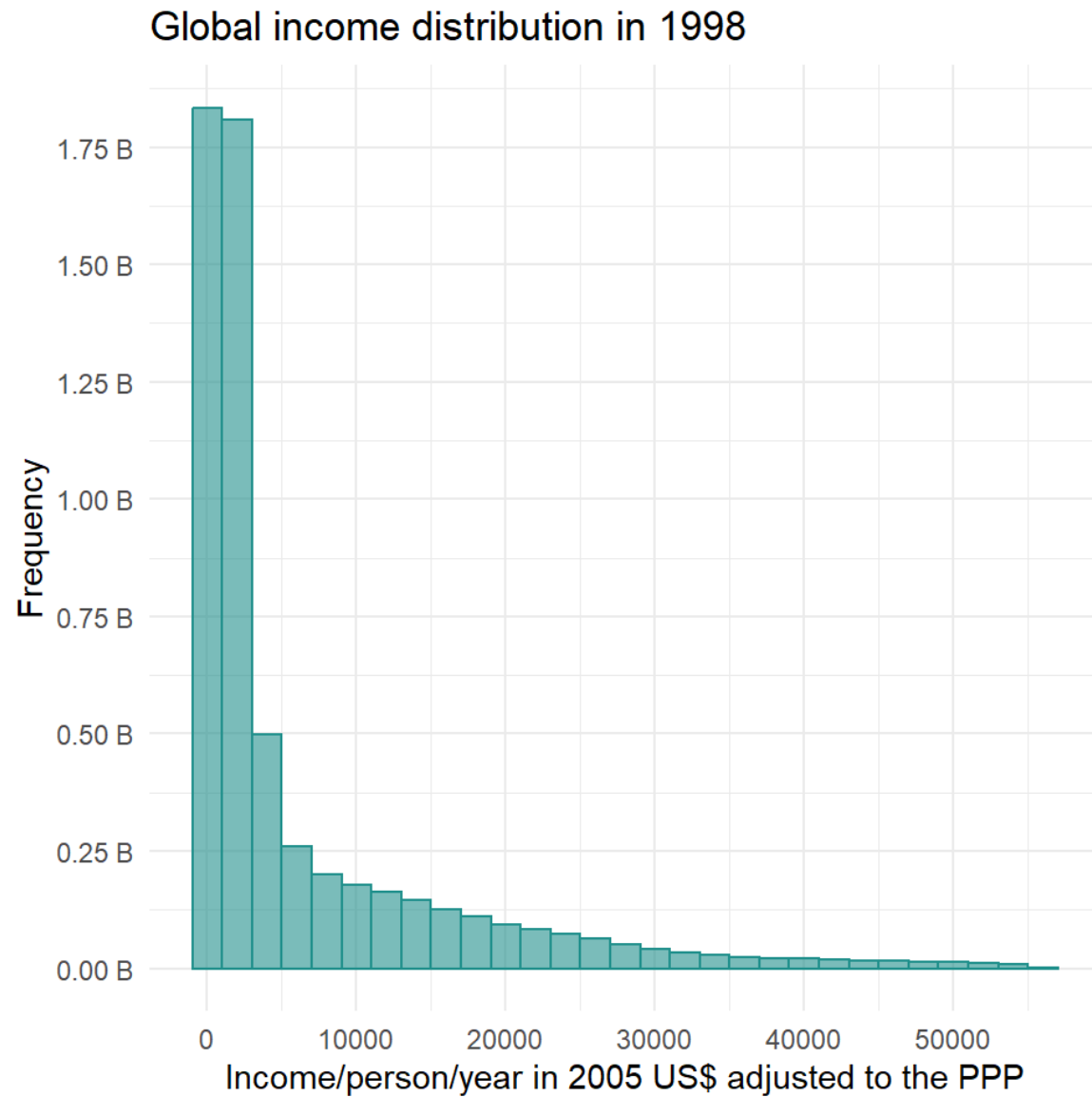


Deciles: Gamma distribution



Example: income deciles and robust-vs-non summary

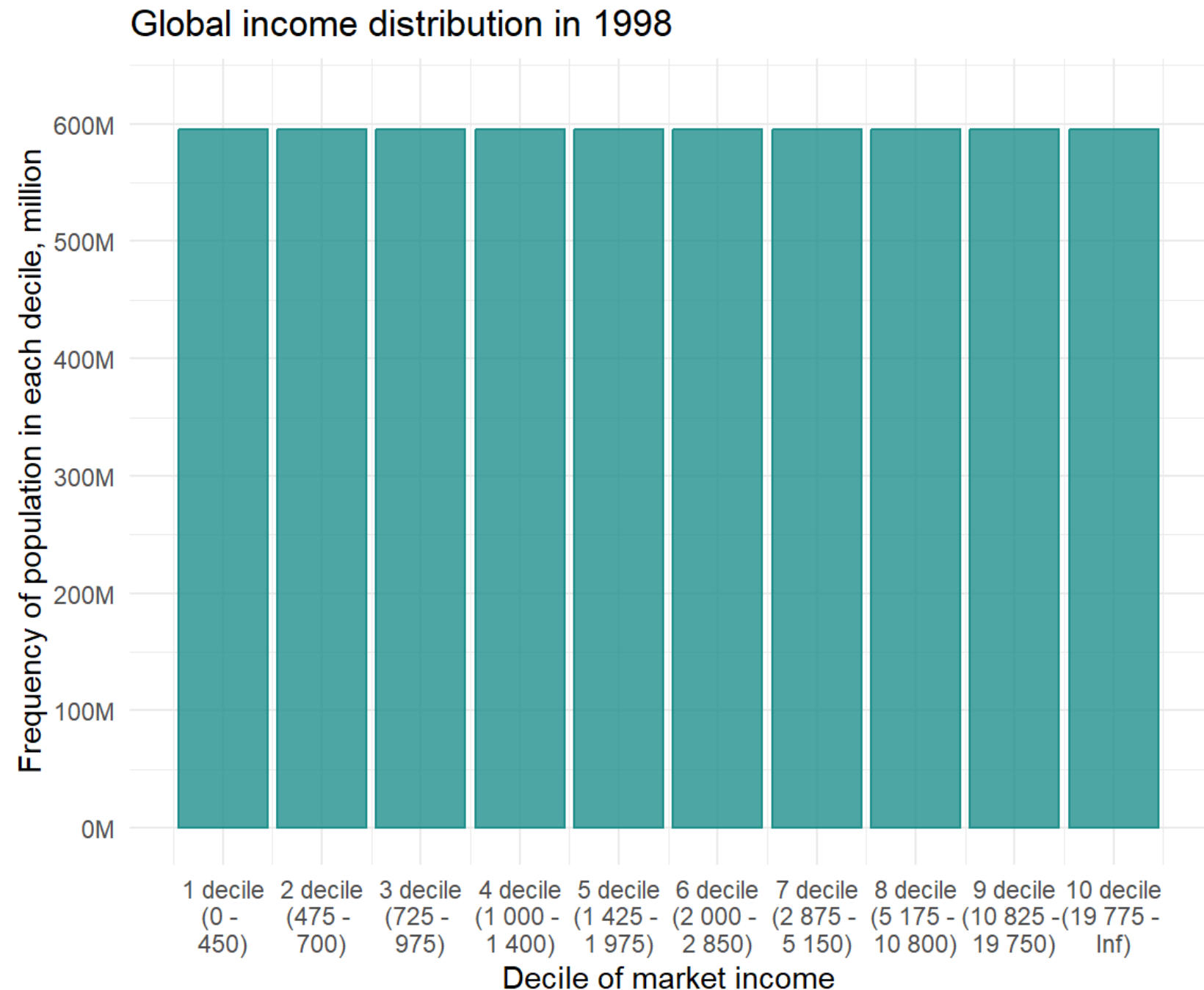
Example: Deciles (1) of Income Distribution



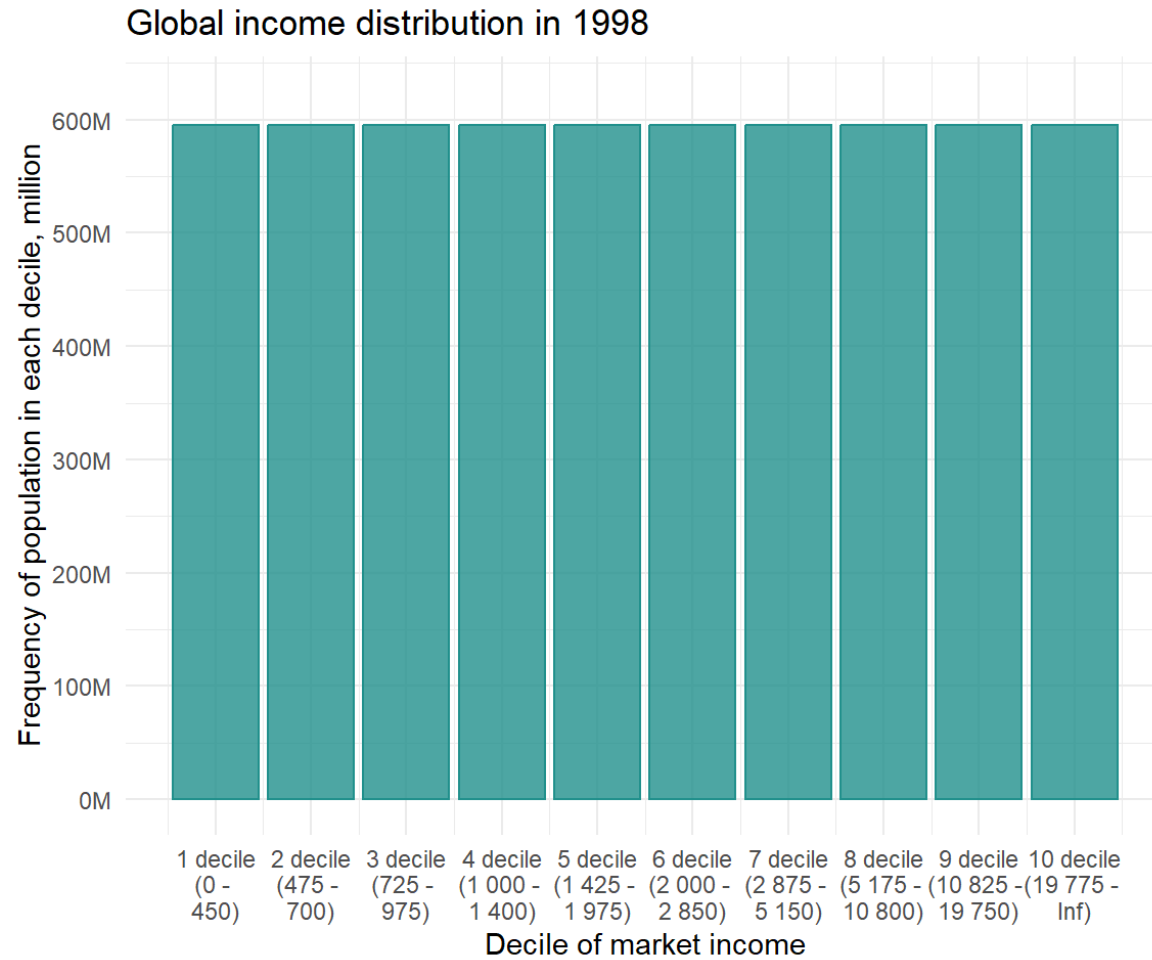
Example: Deciles (2) of Income Distribution

Is the histograms with **bins by deciles** useful?

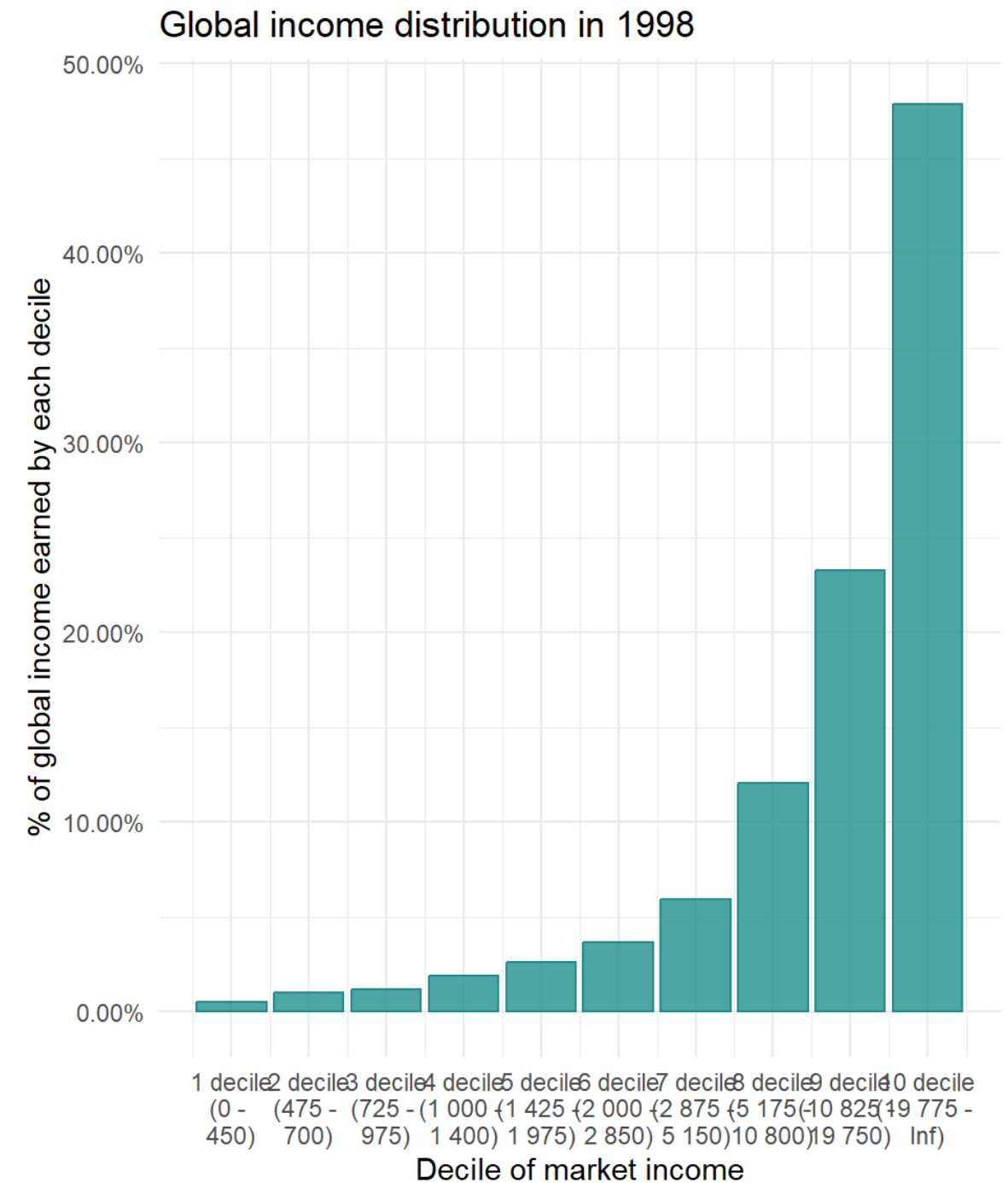
- Let us draw it...
- the total population in 1998 was 6 billion



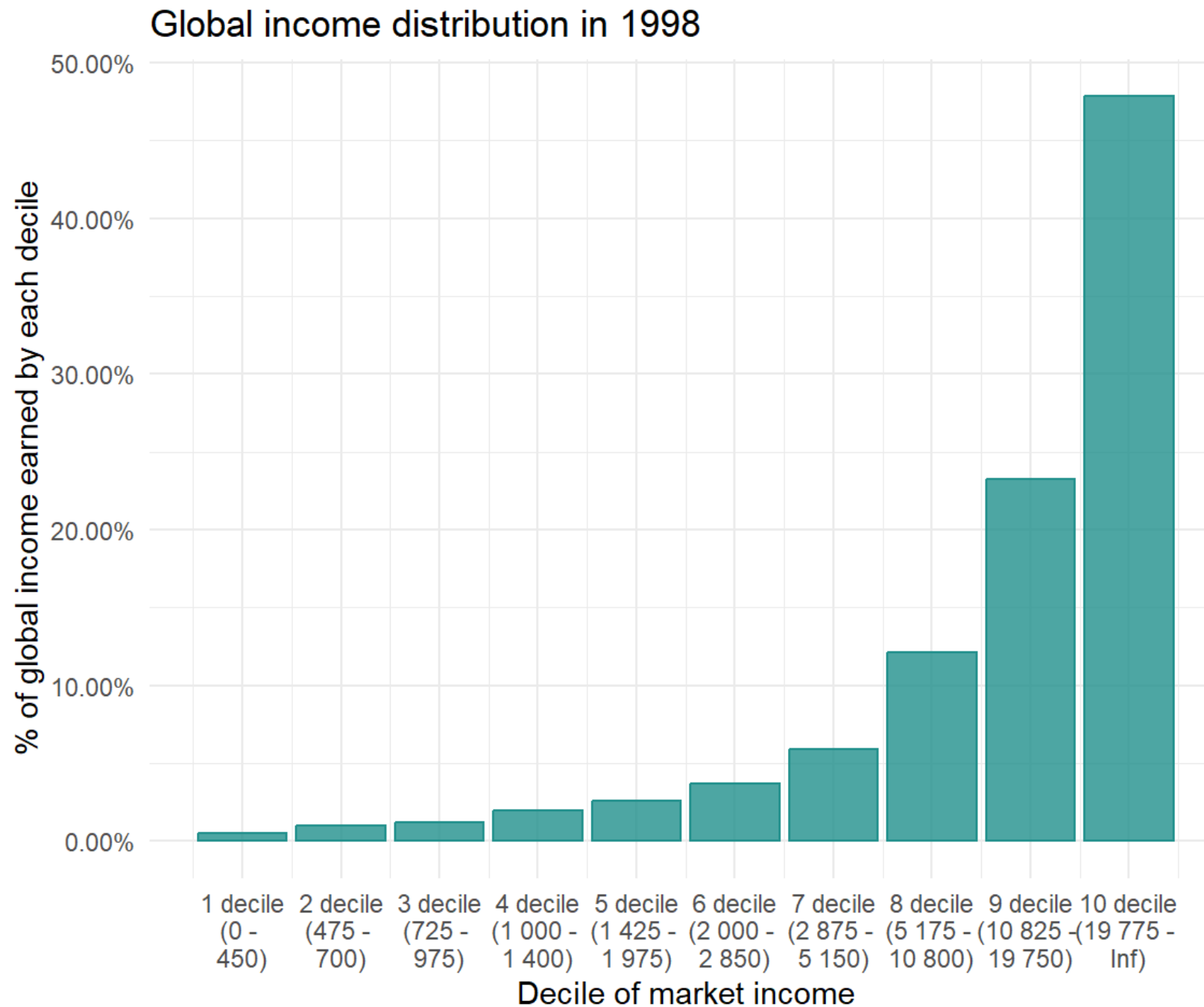
Example: Deciles (3). New insights



Let us plot % share of total income on Y axis.

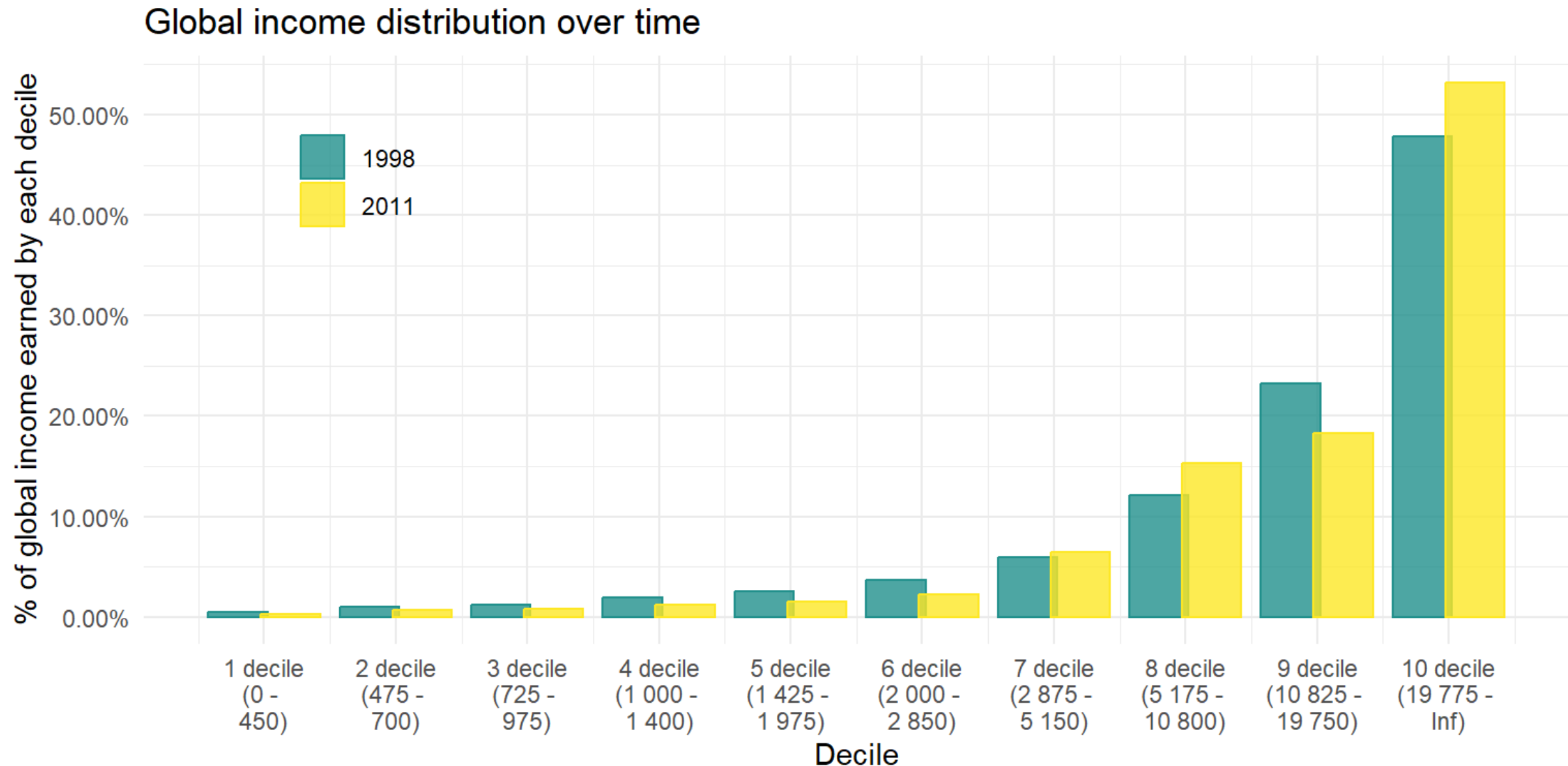


Example: Deciles (4). New insights



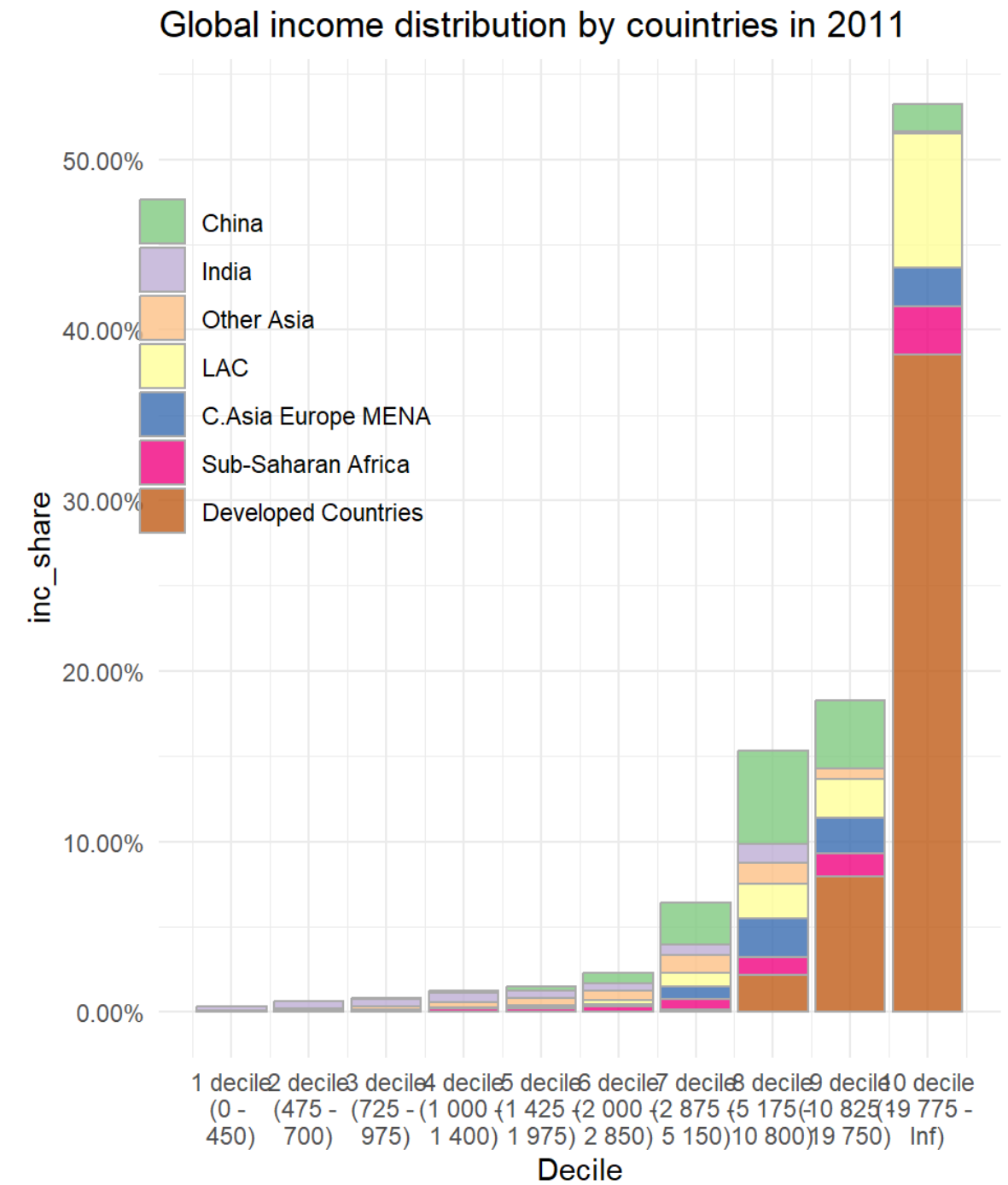
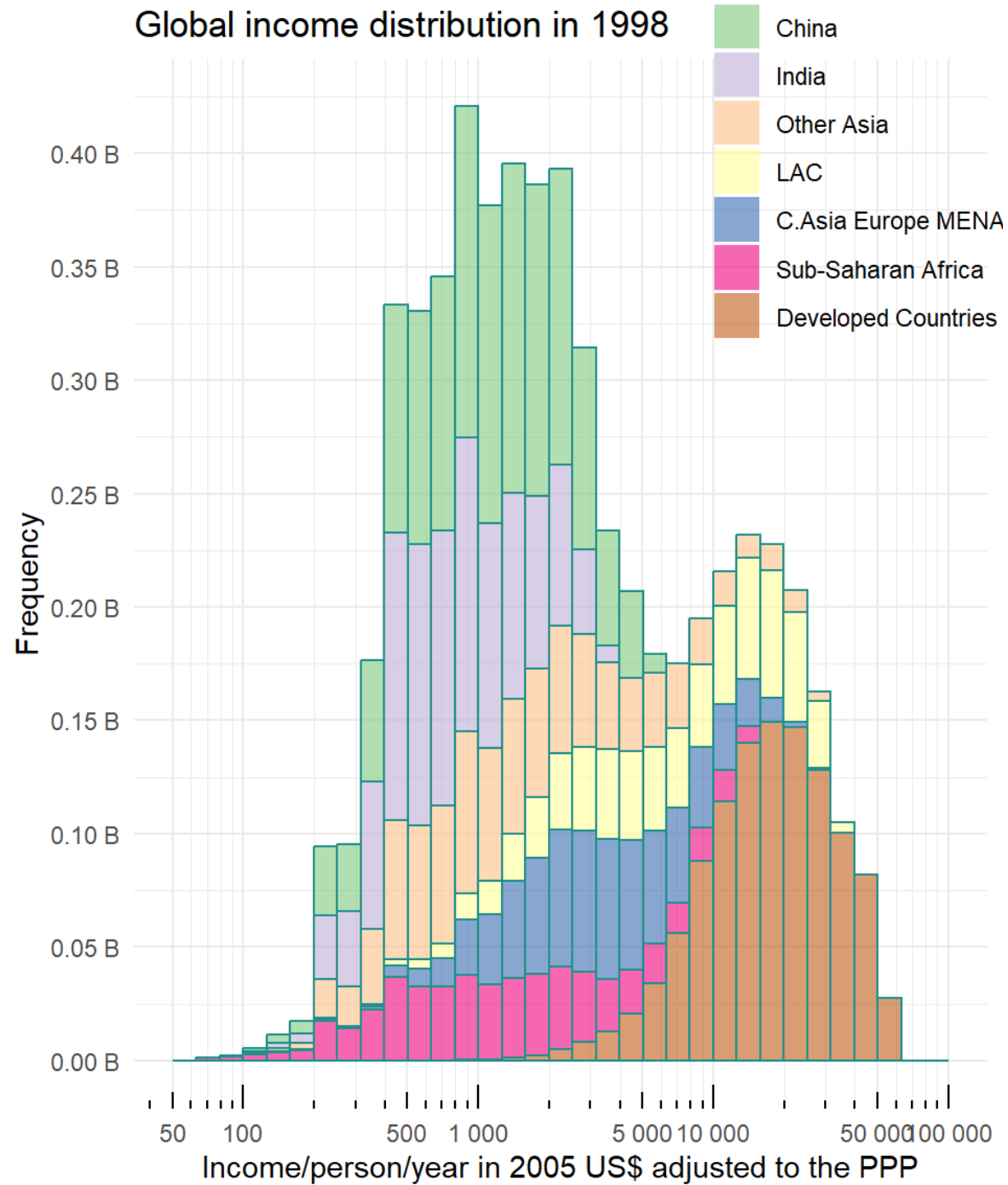
What does it show?
Where is the global income generated and distributed?

Example: Deciles (5). Change over time

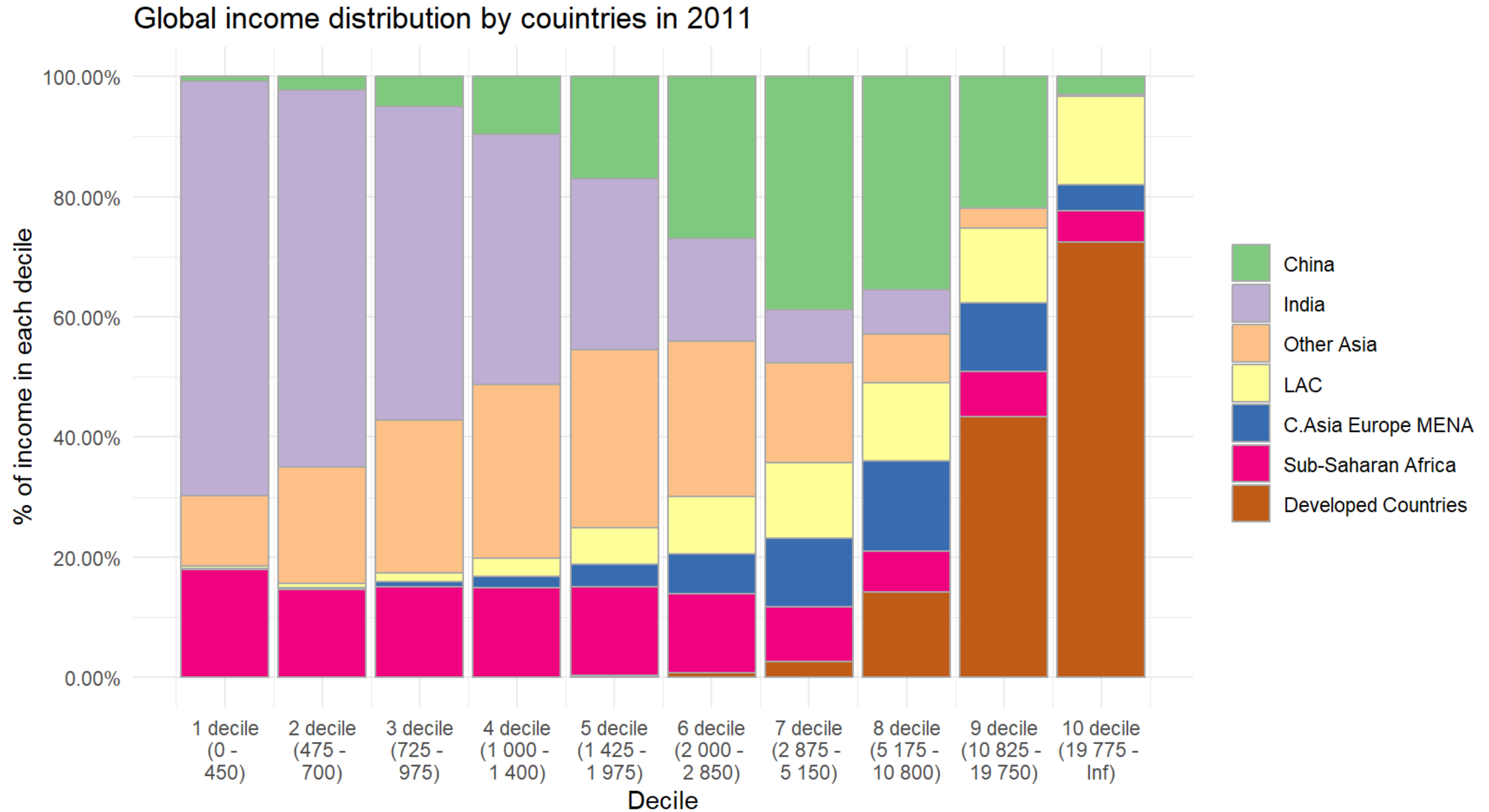


What had changed between 1998 and 2011?

Example: Deciles (6). By country



Example: Deciles (7). By country



Example: inequality, taxes and subsidies (1)

Fiscal policy has a redistribution effect.

- Taxes can accumulate wealth from the “rich”:
 - Direct taxed: income tax
 - Indirect taxed: VAT, and excises
- Subsidies could distribute this wealth to the poor:
 - Direct subsidies: unemployment benefits, child support
 - In-kind subsidies: health and education

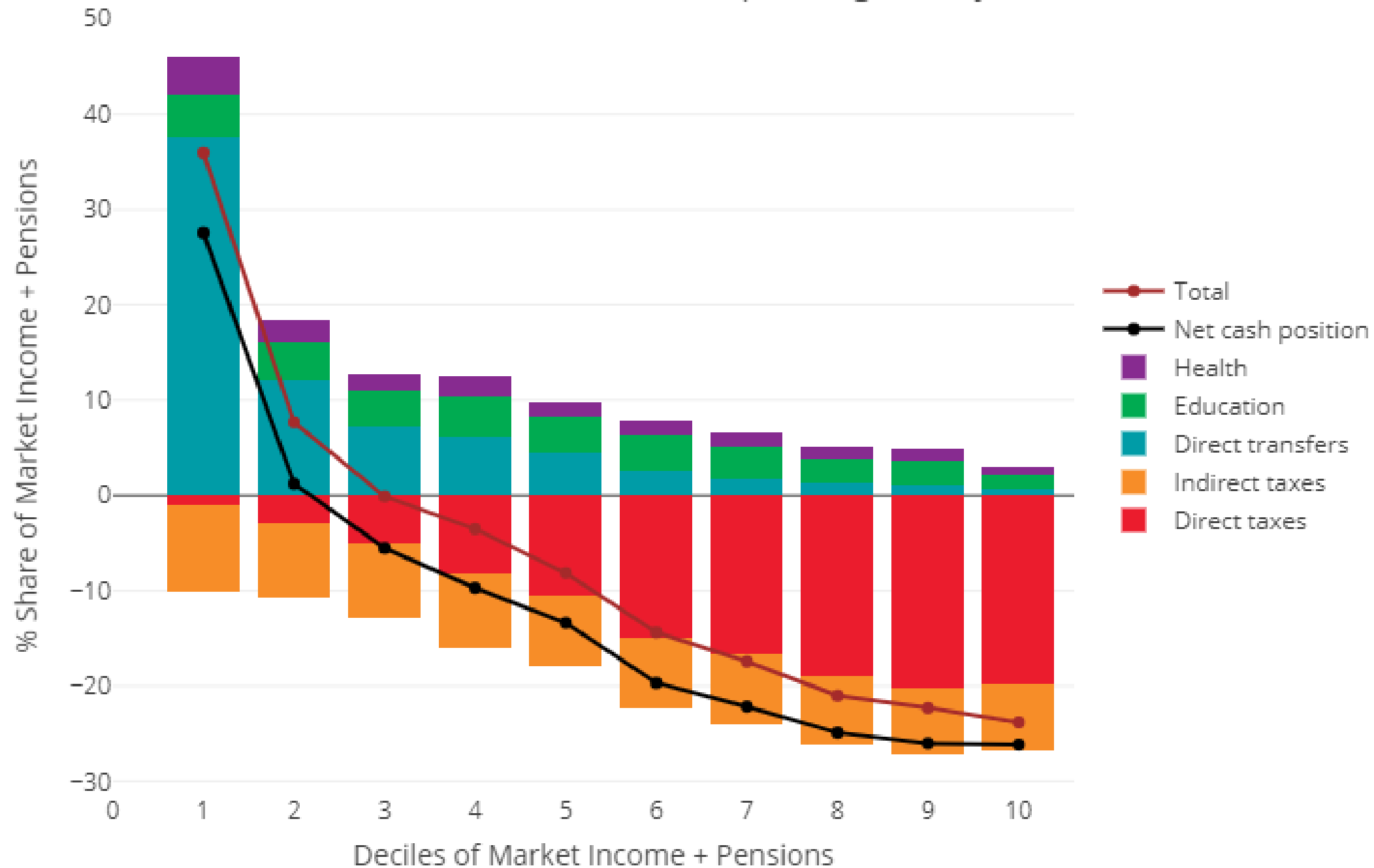
The Data

Data below is produced based on the representative HH surveys.

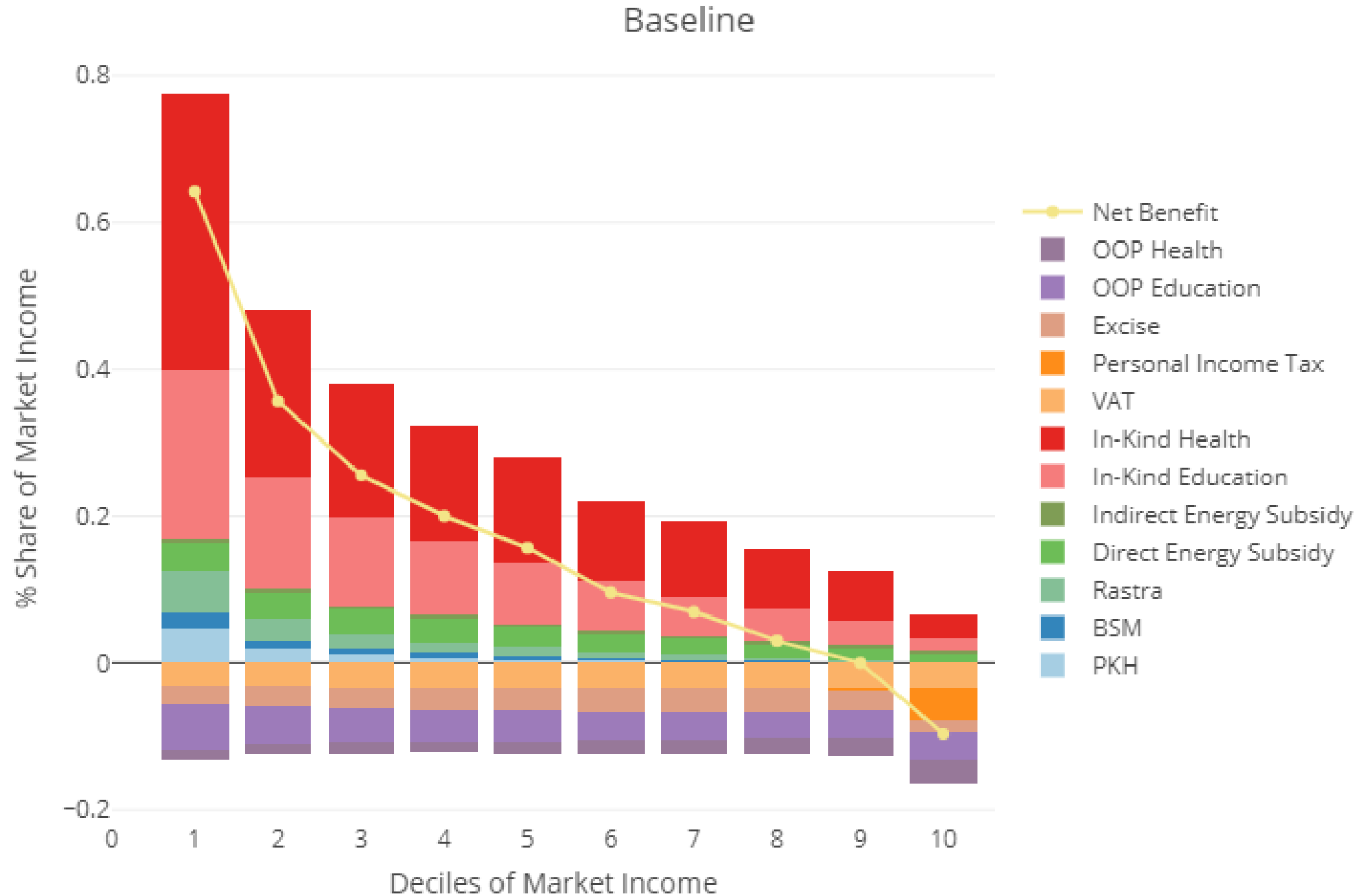
- For each individual income by sources and expenditure by directions was calculated.
- Then individuals were aggregated into the deciles of income.

Example: Taxes (2). Progressive taxation

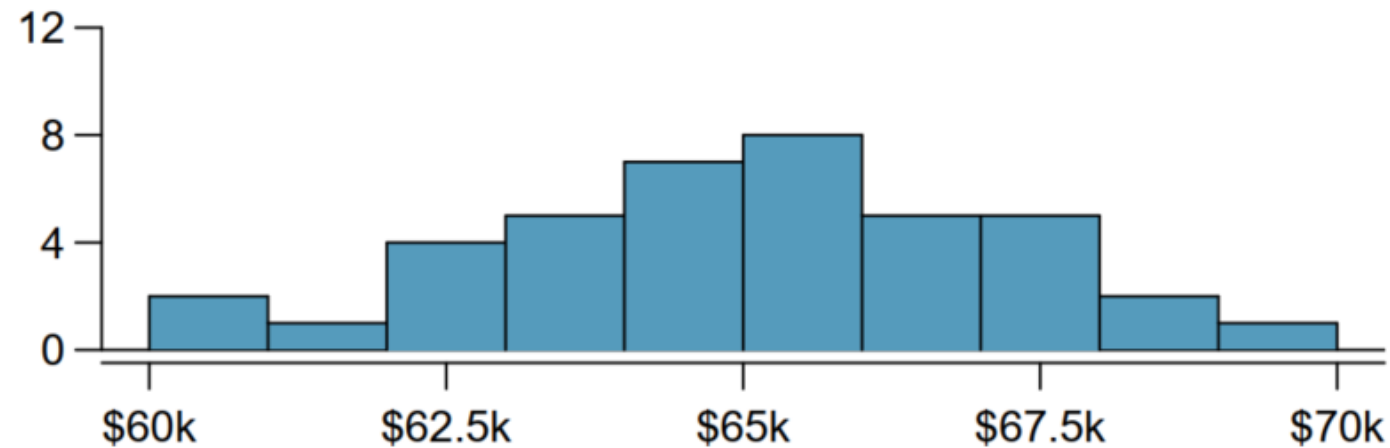
Incidence of taxes and social spending - Policy choice 1



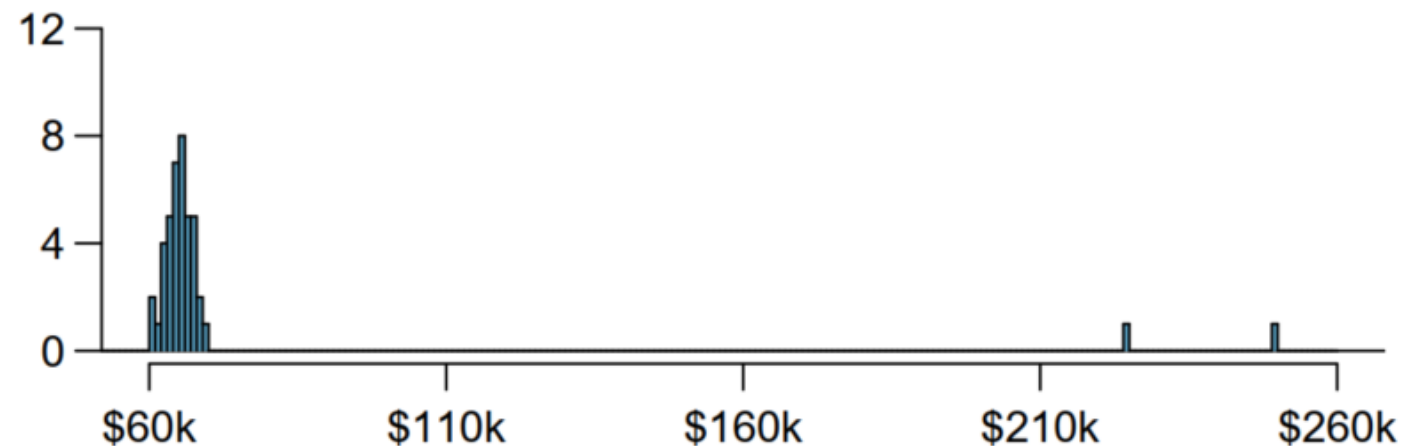
Example: Taxes (3). Regressive taxation



At home: Robust vs non robust variance



(1)

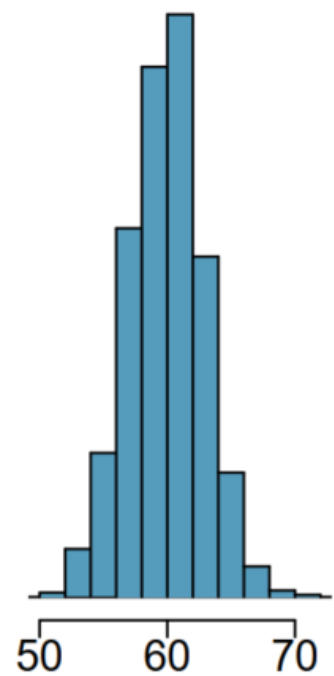


(2)

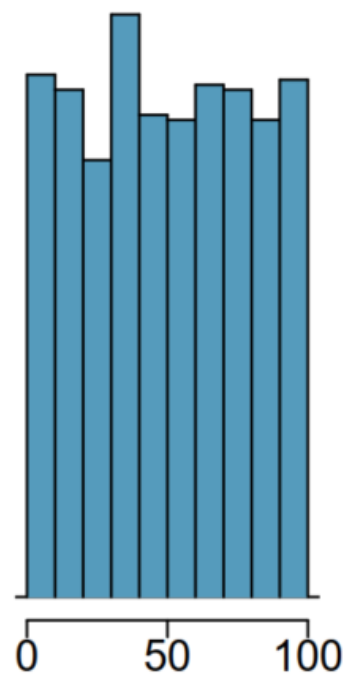
	(1)	(2)
n	40	42
Min.	60,680	60,680
1st Qu.	63,620	63,710
Median	65,240	65,350
Mean	65,090	73,300
3rd Qu.	66,160	66,540
Max.	69,890	250,000
SD	2,122	37,321

- Would the mean or the median best represent the central tendency?
- What does this say about the robustness of the two measures?
- Would the standard deviation or the IQR best represent the amount of variability in the incomes?

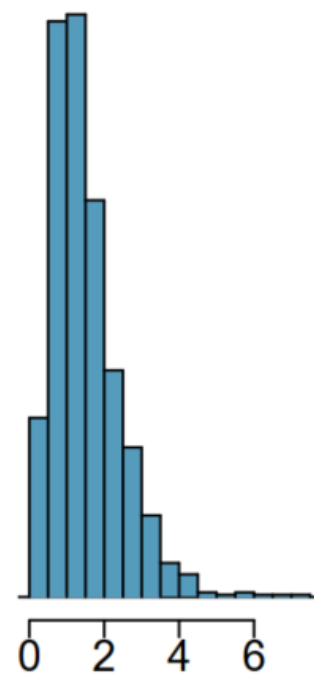
At home: Match distribution with the box plot



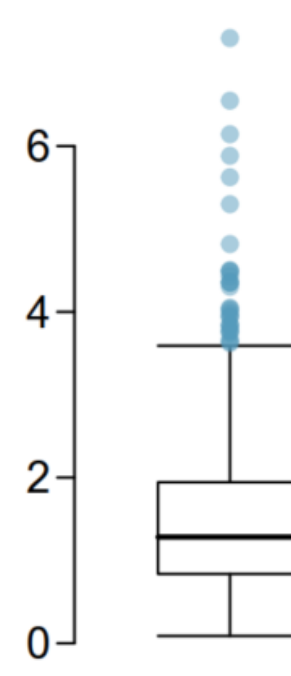
(a)



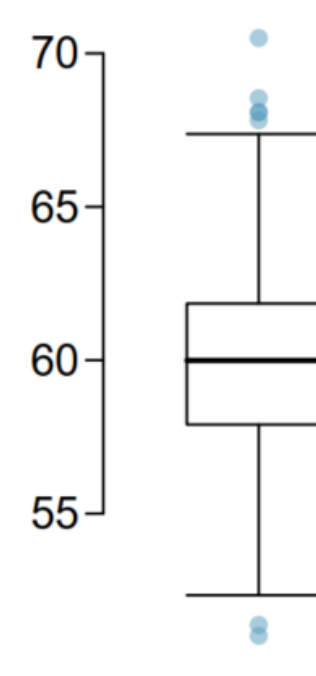
(b)



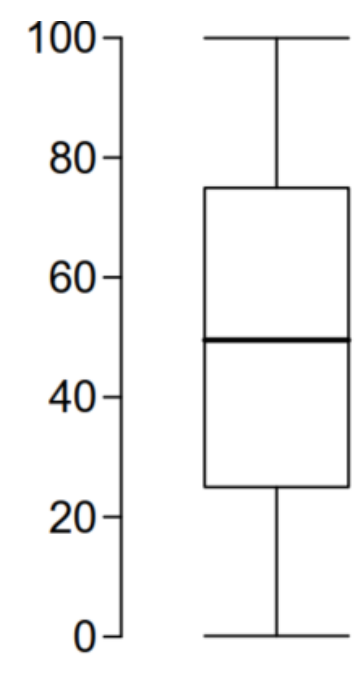
(c)



(1)



(2)



(3)

- Describe the distribution in the histograms below and match them to the box plots.

Application of variance in the descriptive analysis

Example: Measuring stunting

How to measure if a child is stunted?



Stunting is the impaired growth and development that children experience from poor nutrition, repeated infection, and inadequate psycho-social stimulation

- Stunting in the first 1000 days of life has negative consequences on the child's physical and mental development.
- See: [WHO. Stunting in a nutshell](#) and more information on [Hunger and Undernourishment](#).

Global Stunting



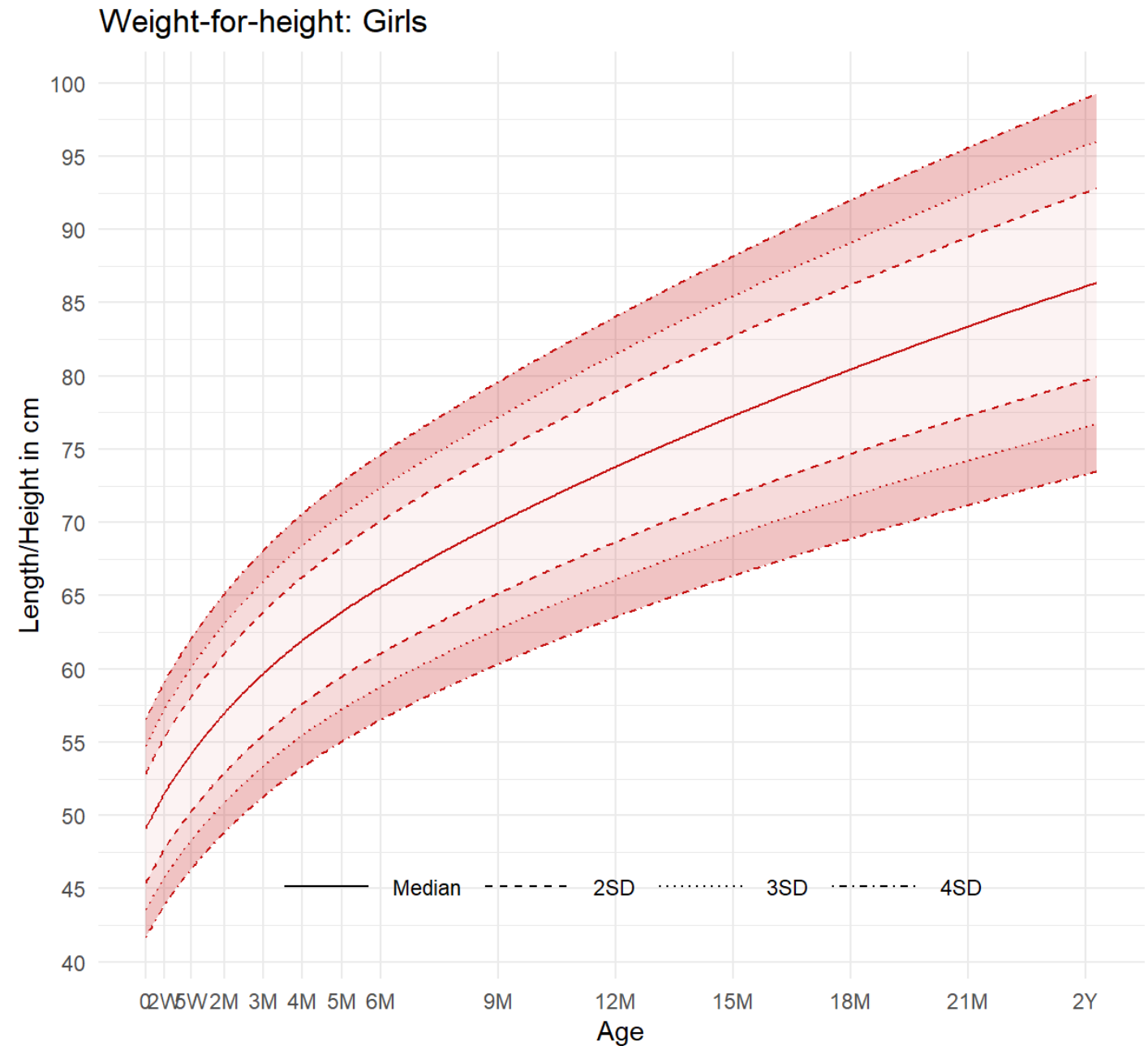
How to measure if the child is stunted?

WHO has developed:

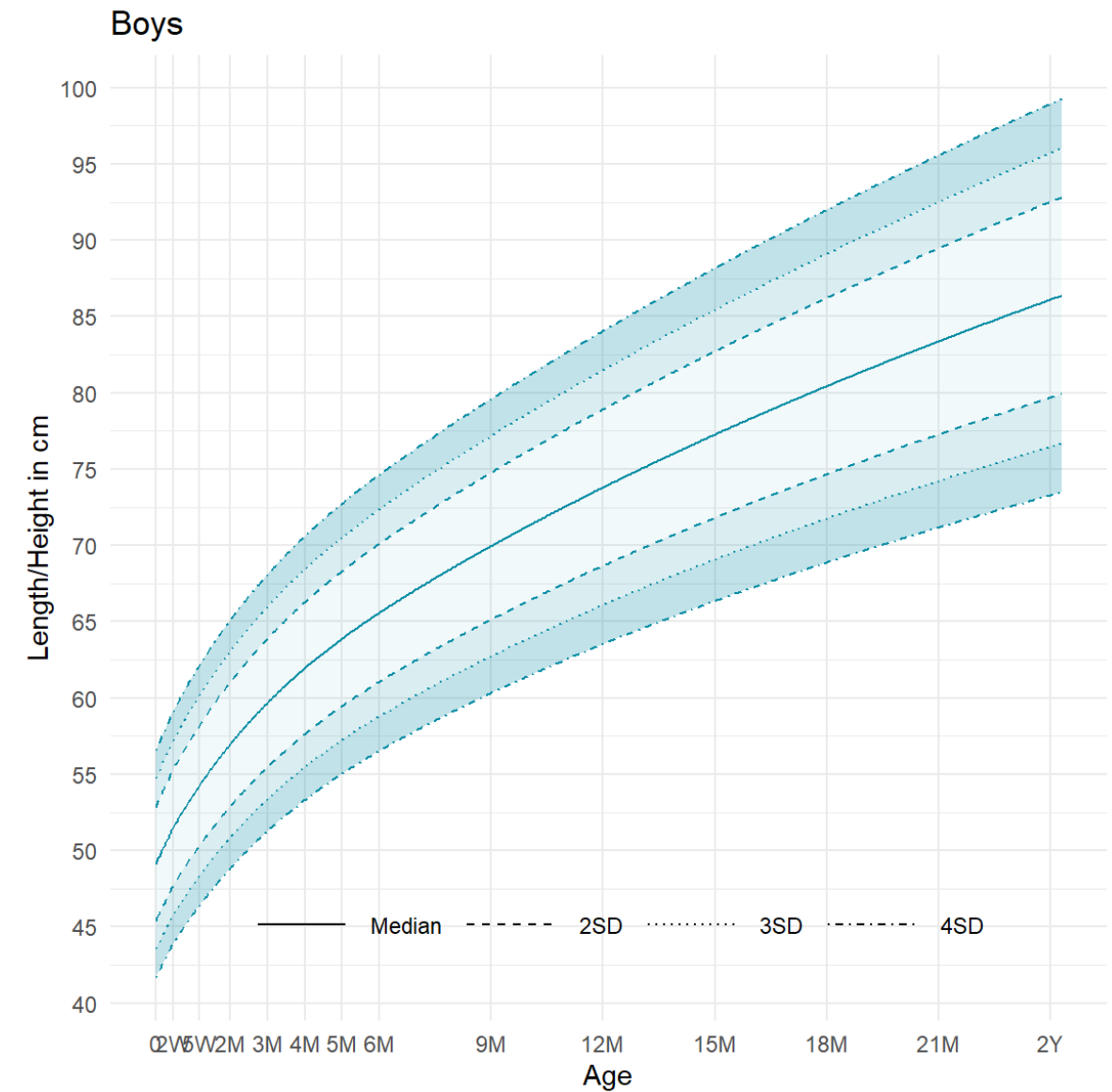
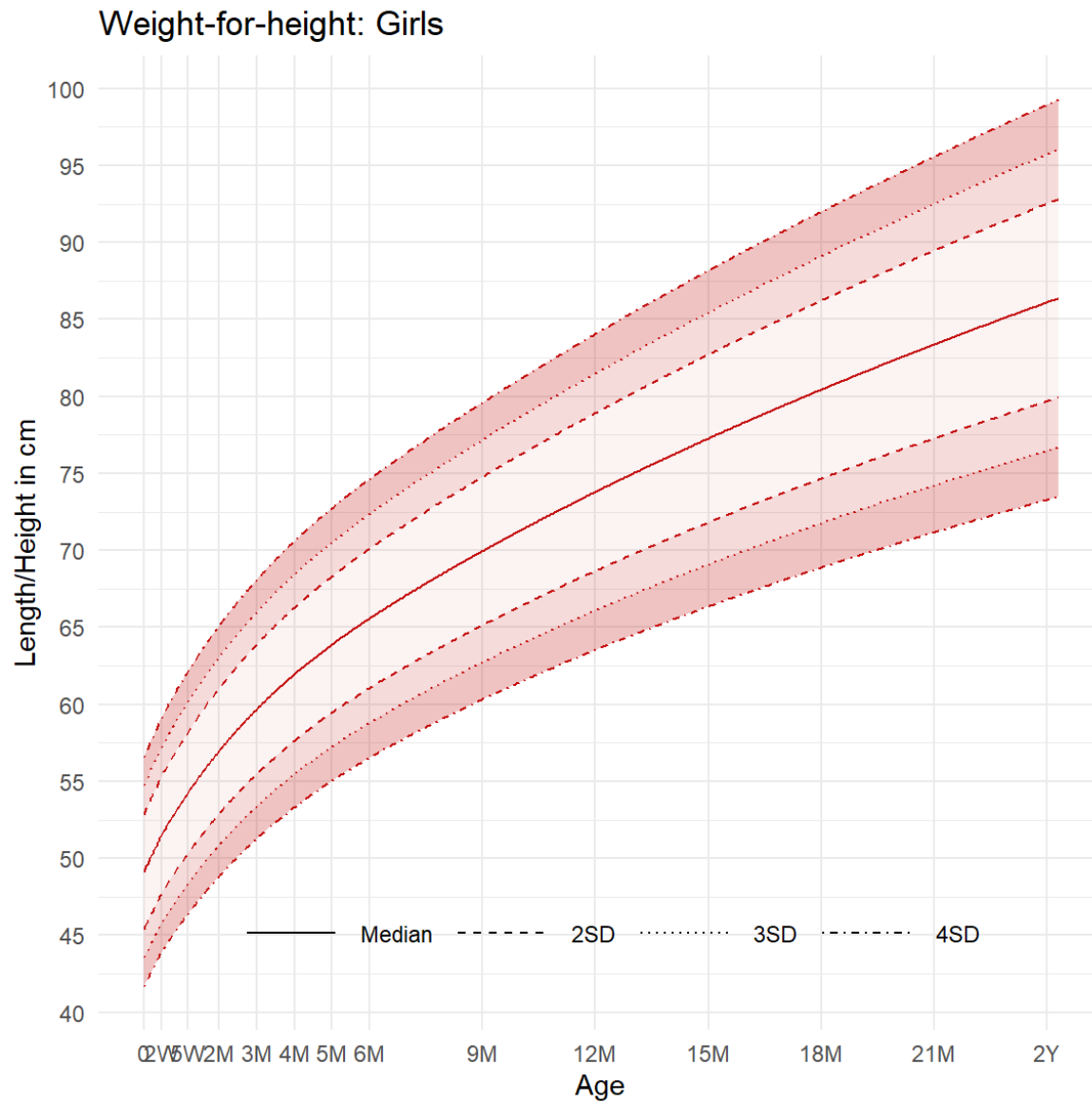
- Child Growth standards

The **Height-for-age** standard stipulates:

- Lower boundaries of heights without stunting
- at all ages 0 days - 12 years.



Length/Height-for-age in boys and girls



How this standard is made?

WHO's data collection:

Site	N	Compliant ^a		
		Boys	Girls	Total
Brazil	309	29	37	66
Ghana	328	103	124	227
India	301	84	89	173
Norway	300	75	73	148
Oman	291	73	76	149
USA	208	64	55	119
All	1737	428	454	882

WHO Child Growth Standards

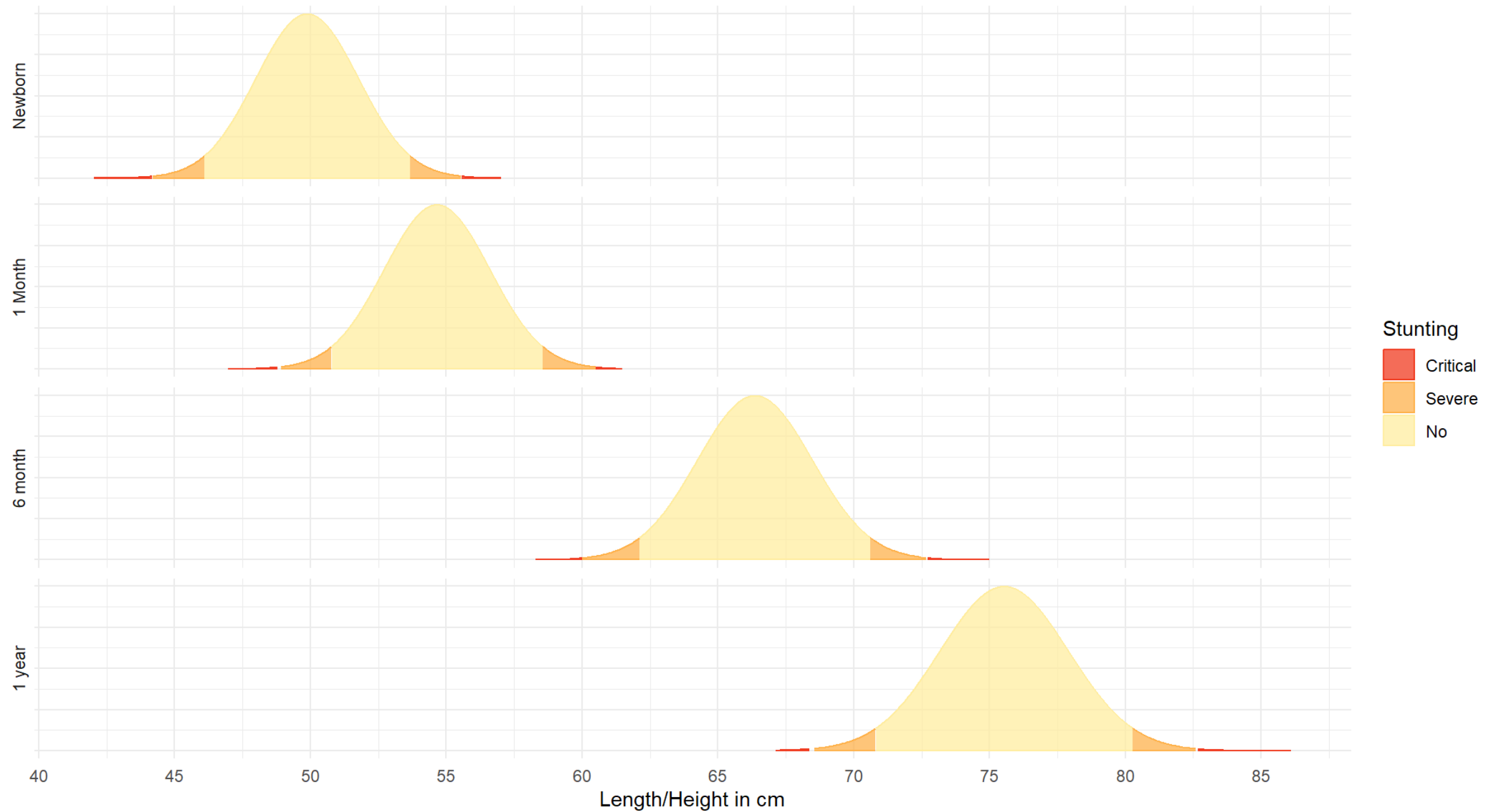
- Random sample of children was followed in each country for several years.
- Children were regularly screened and measured.

Simplified methodology:

For every age group:

- Mean/median and standard deviation were calculated.
- Any child with height-for-age within two standard deviations from the mean is considered non-stunting.
- For a regular population, 95% of children are not stunted.

How this standard is made?



The standard summary

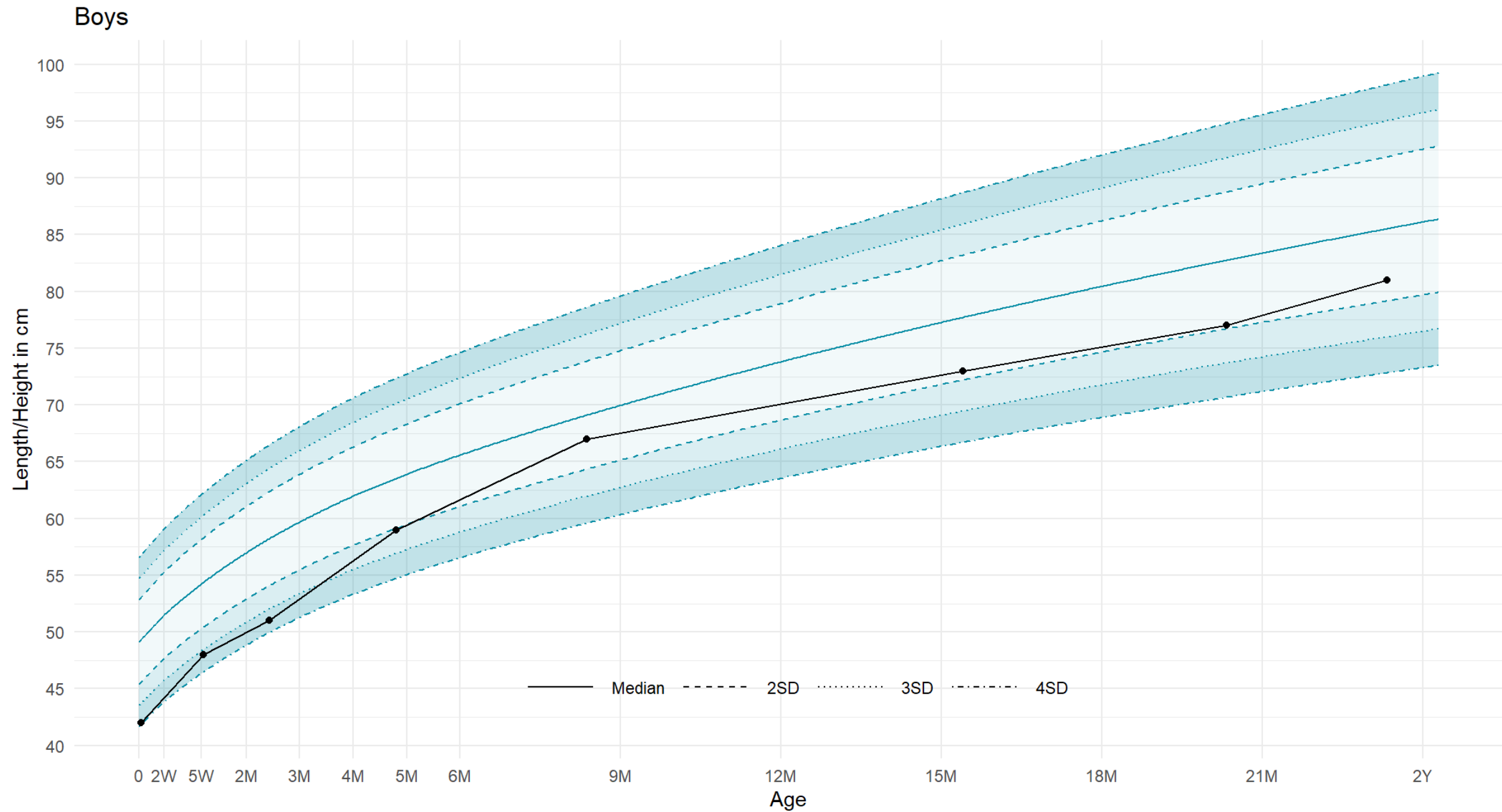
Practically, WHO's data collected on the healthy population helped defining

- probability distribution of **heights** for age and gender of children.
- Mean/Median height
- Standard deviation of height
- The distribution is normal

Stunting was defined as:

- Having actual height 2 standard deviation lower than the central tendency in the healthy population.

Application: let us follow a boy's development:



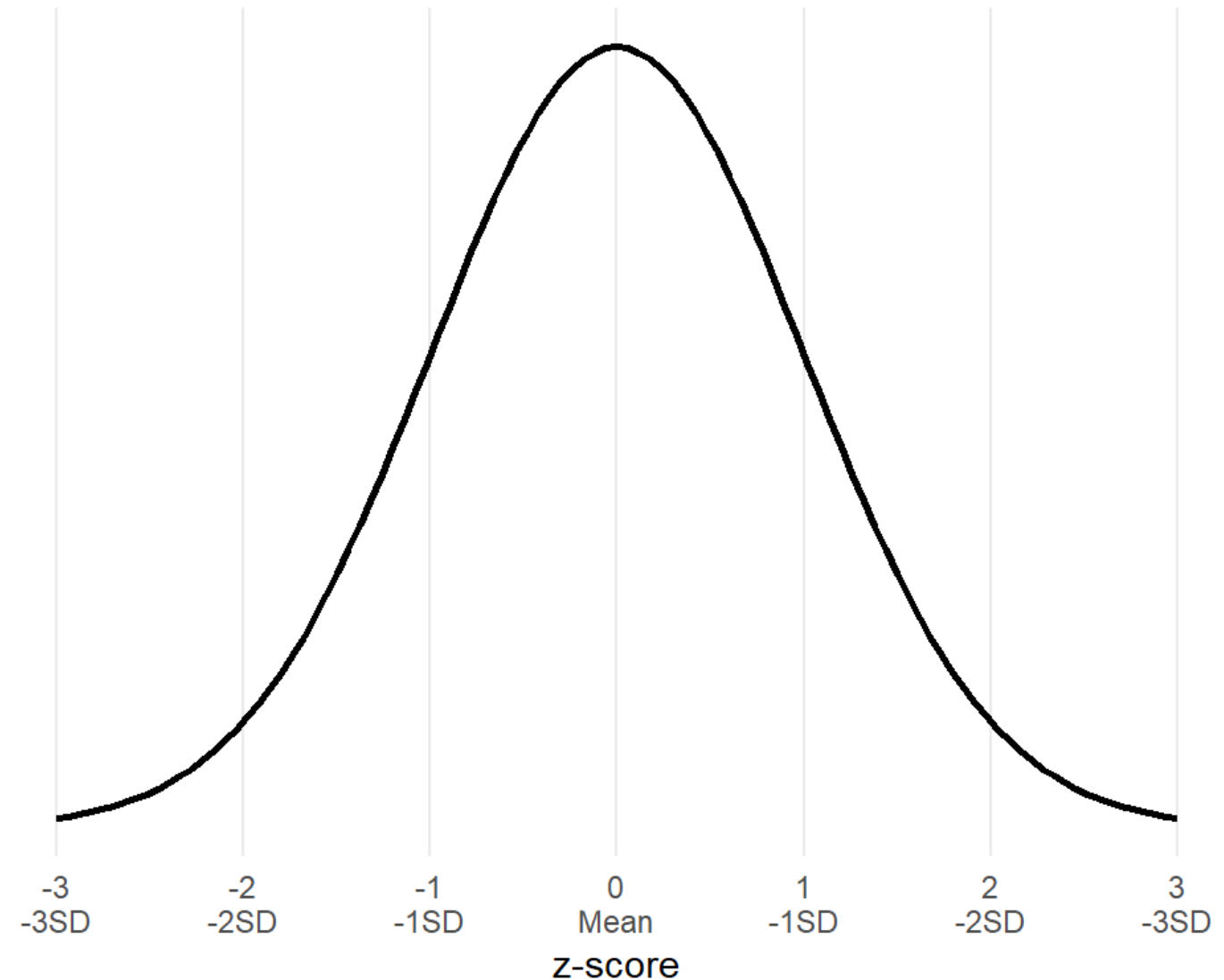
Exam questions on variance

- What is variance and how is it different from the standard deviation.
- When the mean and the standard deviation are not the best measures of respectively the central tendency and the variance?
 - What are the alternative methods?
- Assume that the distribution is normal. What share of observations falls within two SD of mean?
- How the quantiles are constructed?
- Is median one of the quantiles or this is the central tendency?
- What robust measures of variance do you know?

The (Standard) Normal Distribution

Is the symmetric, bell-shaped normal distribution, with:

- $\mu = 0$,
- $SD = 1$.
- X axis contains z-scores.



Z-score

The Z-score of an observation x is the **number of standard deviations** s that it (x) falls above or below the mean $\bar{x}$.

We can compute a z-score of every observation x in our sample. For that we need:

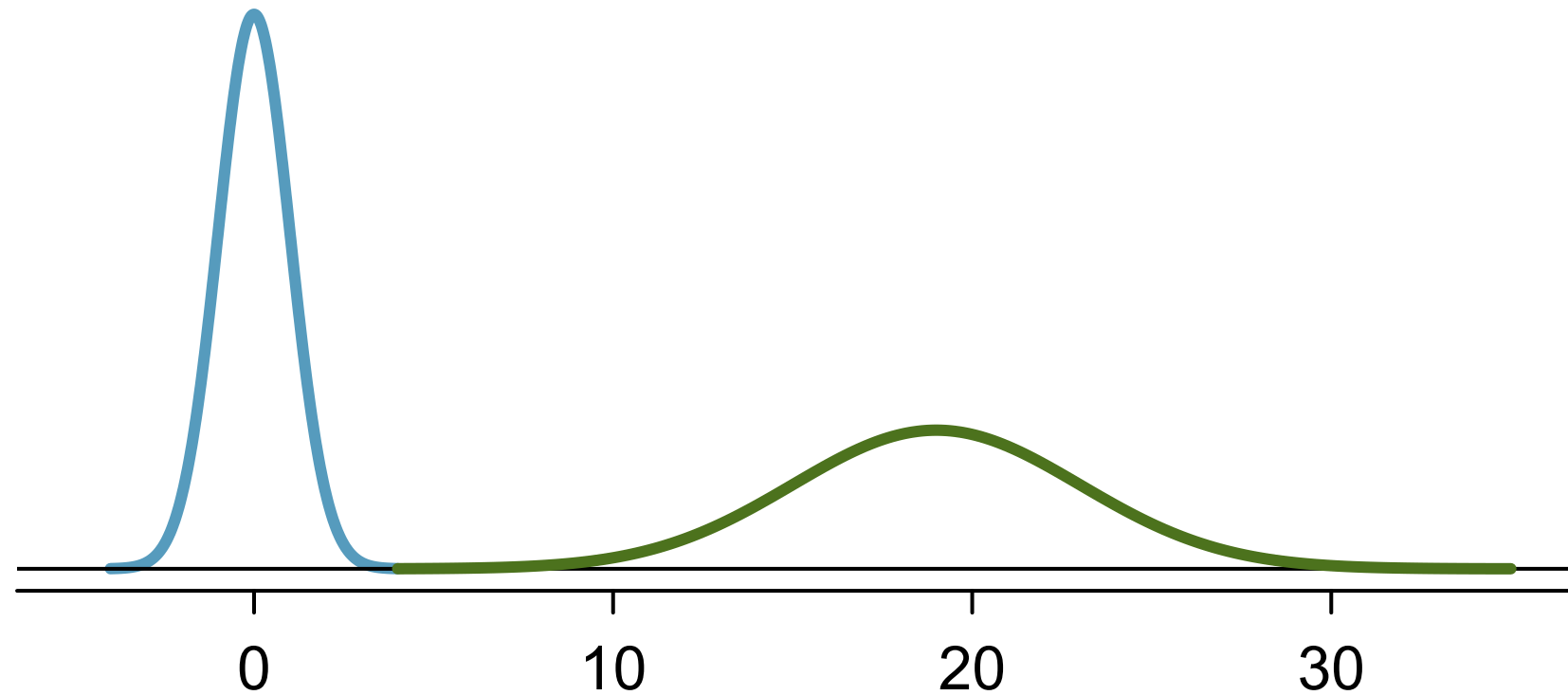
- sample mean $\bar{x}$, and
- sample standard deviation, s .

$$Z = \frac{x - \bar{x}}{s}$$

Please memorize it!

Why do we need it?

Often, we need to compare distributions or locate individual observations on distributions.



This is difficult, when values are expressed in the absolute terms.

Standardization

The process of converting any value of any distribution to the standard normal distribution.

$$Z = \frac{x - \bar{x}}{s}$$

where:

- $\bar{x}$ is the mean, and
- s is the SD, and
- x is the observation.

Example (exam): det. stunting status (1)

Stunting is defined as having heights **2 standard deviations lower** than the average height of a healthy child of the same gender and age.

We've measured that:

- a 5 years old girl
- has heights of 97 cm.

Can we consider her suffering from stunting?

According to the **Child Growth Standards** about **height-for-age**, the height of a healthy girl at the age of 5 years old:

- is normally distributed with
- the mean $\mu = 109$ cm and
- the SD $s = 4.73$ cm.

Example (exam): det. stunting status (2)

Computing the z-score of the observation:

$$Z = \frac{97 - 109}{4.73} = \frac{-12}{4.73} = -2.54$$

Conclusion:

- Observed height of a 5 y.o. girl falls below -2 standard deviations.
- Thus she might be suffering from stunting if other causes of growth deprevation (e.g. genetic) are eliminated.

Example 2 (exam) DIY

1. The boy's height is 76 cm. He is 1.5 Years old.
 - Is this boy stunted given that the expected height in the corresponding population of that age is 80.3 and the SD is 2.6?
 - Calculate corresponding statistics and write a brief conclusion.
2. The girl's weight is 13.8 kg. She is 5 years old.
 - Does this girl suffer from waisting?
 - Expected weight in the corresponding population of that age is 18.3 and the SD is 2.05?
 - Calculate corresponding statistics and write a brief conclusion.

Exam questions on the Normal distribution

- What is the difference between a normal distribution of age and a standard normal distribution?
- What is the z-score?

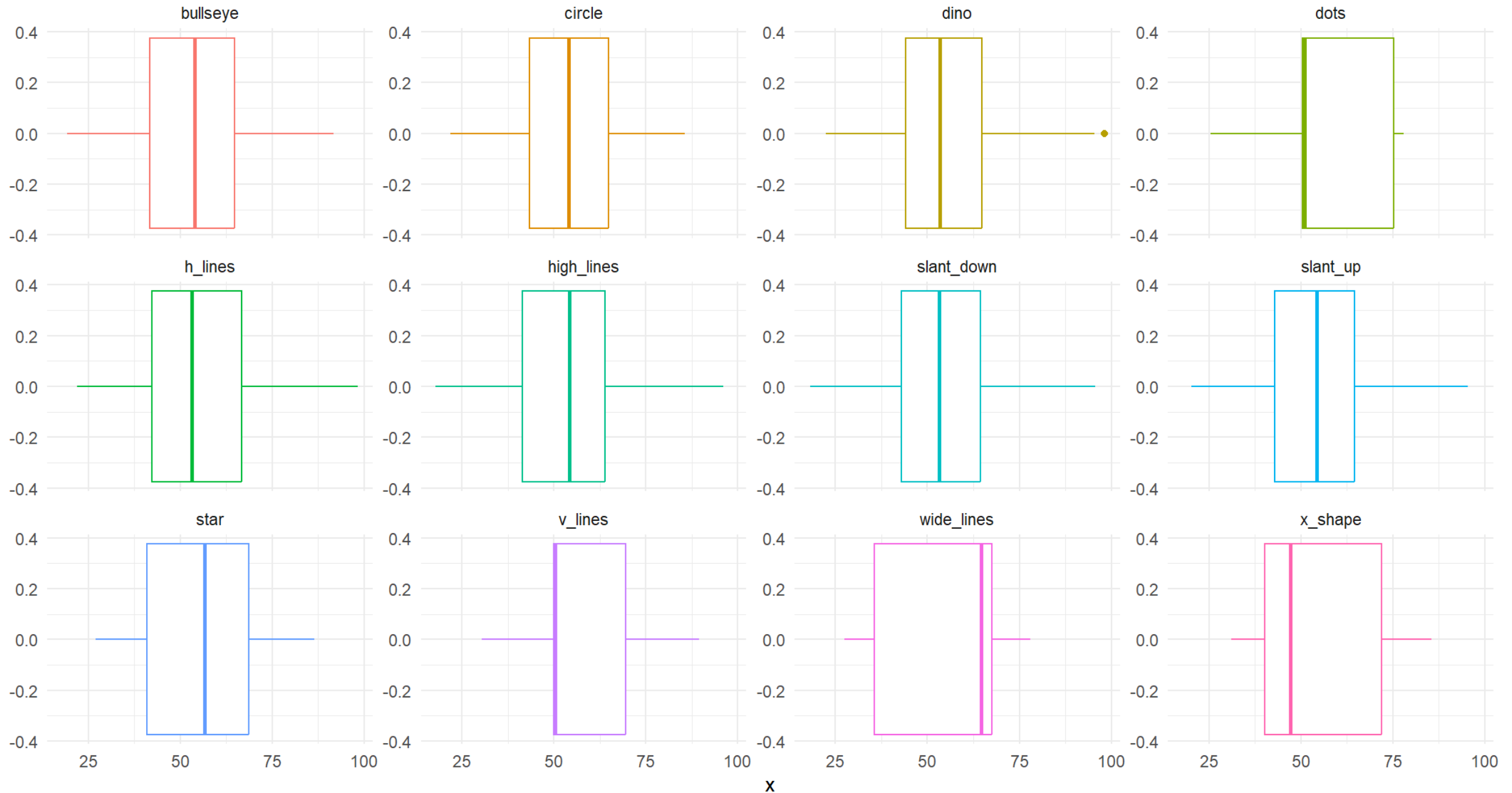
When univariate data analysis is not enough?

Datasaur: Synthetic data (1)

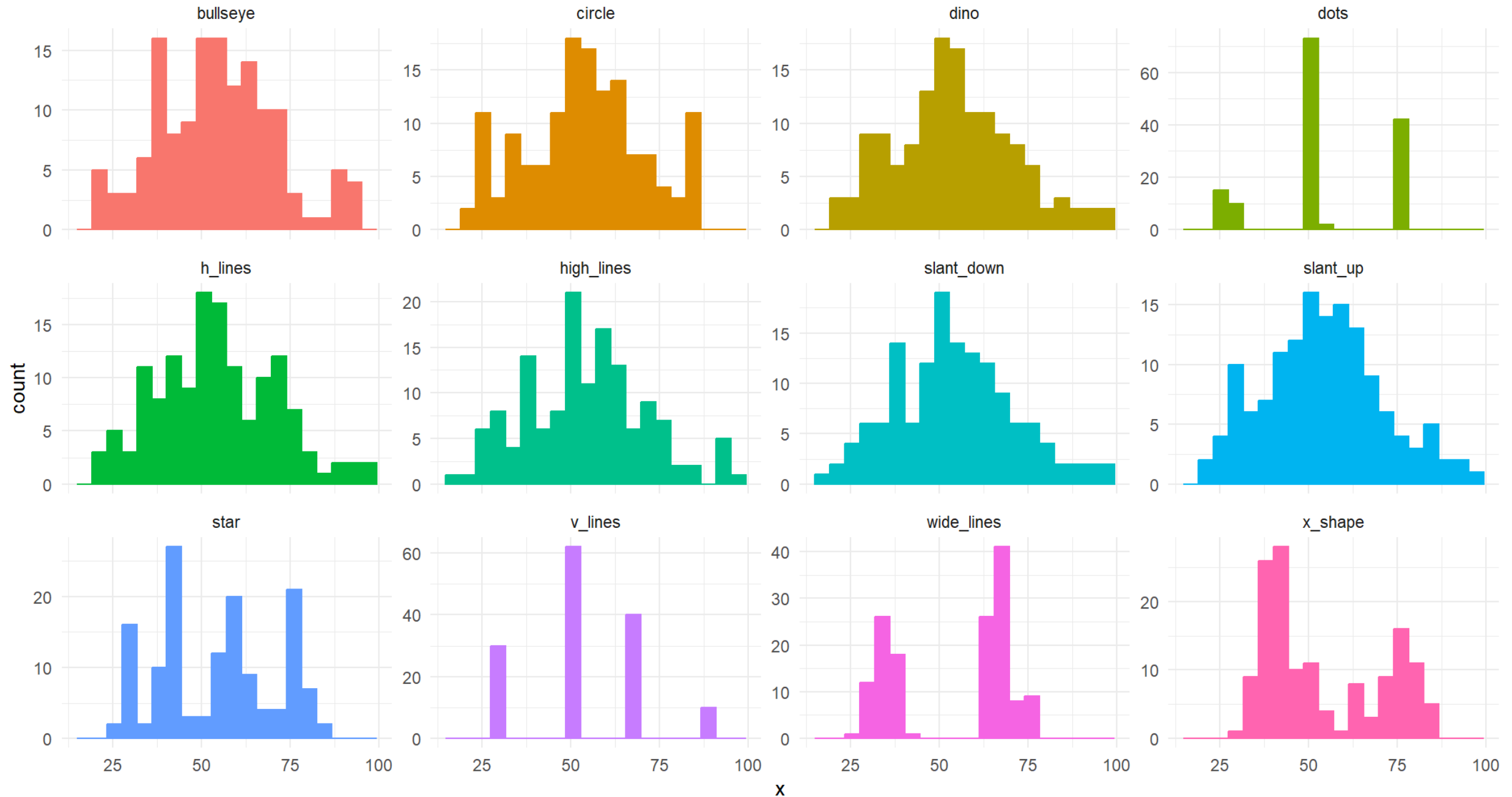
- There are 13 synthetic data sets that produce similar descriptive statistics and distributions:

dataset	mean_x	mean_y
away	54.27	47.83
bullseye	54.27	47.83
circle	54.27	47.84
dino	54.26	47.83
dots	54.26	47.84
h_lines	54.26	47.83
high_lines	54.27	47.84
slant_down	54.27	47.84
slant_up	54.27	47.83
star	54.27	47.84
v_lines	54.27	47.84
wide_lines	54.27	47.83
x_shape	54.26	47.84

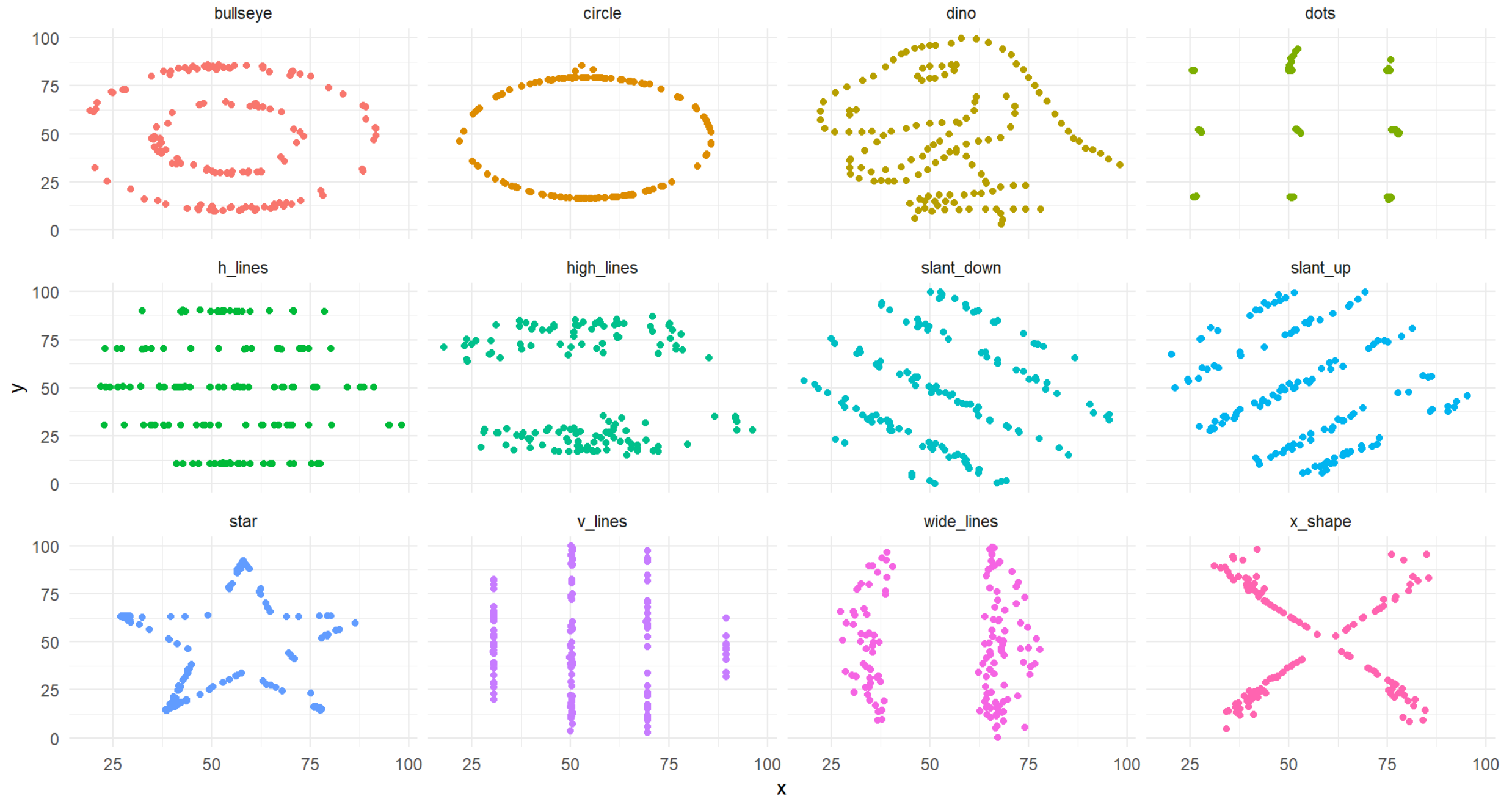
Univariate analysis of the Datasaur (1)



Univariate analysis of the Datasaur (2)



Bi-variate distribution is important!



Takeaways checklist

Read more about income inequality at **Our World in Data**

- The history of global economic inequality
- Global Inequality of Opportunity
- Global economic inequality: what matters most for your living conditions is not who you are, but where you are
- Income inequality before and after taxes: How much do countries redistribute income?
- OWID Data Collection: Inequality and Poverty
- How has inequality in the United Kingdom changed over the long run?
- How unequal were pre-industrial societies?
- How has income inequality within countries evolved over the past century?

Read also:

Read the paper below **before week 4**: ([Kuchenbecker et al. 2017](#))

- Kuchenbecker, J., Reinbott, A., Mtimuni, B., Krawinkel, M. B., & Jordan, I. (2017). Nutrition education improves dietary diversity of children 6-23 months at community-level: Results from a cluster randomized controlled trial in Malawi. PLOS ONE, 12, e0175216.
- Available on Ilias.

Key concepts covered in the lecture

Univariate data

The Histogram:

- bars, frequencies, bins and bins width;

The Density plot:

- Density vs frequency;
- How to make a density plot;

The Cumulative distribution:

- How to make it;
- Equal vs unequal distribution;
- Lorenz Curve;

The Variance:

- Robust vs non robust statistical measures of variance;
- Variance and Standard deviation;
- Use of SD;
- Units of measurement in SD;
- 1, 2, 3 SD from the mean/median;

The Normal Distribution:

- Mean and Variance in the Normal distribution;
- Standard normal distribution;
- Z-scores and standardization;

References

- Cairo, Alberto. 2016. “Download the Datasaurus: Never trust summary statistics alone; always visualize your data.” <http://www.thefunctionalart.com/2016/08/download-datasaurus-never-trust-summary.html>.
- Diez, David, Mine Cetinkaya-Rundel, and Christopher D Barr. 2022. *OpenIntro Statistics: Fourth Edition*. OpenIntro. <https://www.openintro.org/book/os/>.
- Dowdy, Shirley, Stanley Weardon, and Daniel Chilko. 2004. *Statistics for Research*. John Wiley & Sons, Inc. <https://doi.org/10.1002/0471477435>.
- Kuchenbecker, Judith, Anika Reinbott, Beatrice Mtimuni, Michael B. Krawinkel, and Irmgard Jordan. 2017. “Nutrition Education Improves Dietary Diversity of Children 6-23 Months at Community-Level: Results from a Cluster Randomized Controlled Trial in Malawi.” Edited by Jacobus P. van Wouwe. *PLOS ONE* 12 (4): e0175216. <https://doi.org/10.1371/journal.pone.0175216>.
- Matejka, Justin, and George Fitzmaurice. 2017. “Same Stats, Different Graphs: Generating Datasets with Varied Appearance and Identical Statistics through Simulated Annealing.”
- Todaro, Michael, and Stephen C. Smith. 2014. *Economic Development*. 11th ed. Pearson.
- Wilke, Claus O. 2019. *Fundamentals of Data Visualization: A Primer on Making Informative and Compelling Figures*. O'Reilly Media.

